Measuring the Gap: Semantic Alignment Between AI-for-Sustainability Research and SDG Policy Frameworks

Contents

Abstract

The Sustainable Development Goals (SDGs) are the dominant international development framework. Shared SDG labels do not necessarily imply shared semantic framing between research and policy. Existing bibliometric studies map AI research to SDGs using keyword or classification methods (Armitage et al., 2020; Kashnitsky et al., 2024), but rarely compare the semantic content of research and policy discourse. This study uses Sentence-BERT embeddings and a nearest-centroid SDG instrument validated against expert-labelled benchmarks (macro-F1 = 0.716) to map semantic distance between 2,543,698 AI-for-sustainability abstracts and 54,082 segments from 2,456 institutional SDG policy sources. The analysis separates coverage gaps (which SDGs each corpus emphasises) from within-SDG semantic gaps (how far texts assigned to the same SDG sit apart). Because the corpora differ in register (Biber and Conrad, 2009), these gaps are observed textual-semantic divergence, not substantive disagreement.

The results find: First, policy discourse concentrates on SDGs 17 (Partnerships, 48.2%), 13 (Climate Action, 25.0%), and 16 (Peace and Strong Institutions, 14.5%), together 87.7% of policy coverage. Second, research concentrates on SDGs 3 (Good Health, 18.5%), 4 (Quality Education, artefact-affected, 14.0%), and 9 (Industry and Infrastructure, 9.7%). Third, the largest within-SDG semantic gaps occur in SDGs 17 (0.671), 13 (0.528), 7 (0.497), 6 (0.491), and 10 (0.432), though the SDG 17 estimate is sensitive to reference-source choice (see Section 5.4). Fourth, research coverage alone does not predict semantic divergence (H1c: $r = 0.040$, $p = 0.879$; $n = 17$) — this dissociation between attention and framing is the study's most robust finding, stable across embedding models and sample sizes. Fifth, H1a, the paper's primary hypothesis, tests whether the coverage gap predicts semantic divergence: this is supported in the canonical specification ($r = 0.604$, $p = 0.010$) but reverses sign under two of four alternative reference centroids (Section 3.8), so it is reported as exploratory, not as a fixed effect. Two further exploratory associations — policy coverage with divergence (H1d, $r = 0.582$, $p = 0.014$) and signed dominance with divergence (H1b, $r = -0.534$, $p = 0.027$) — show the same sign-reversal pattern and carry the same caveat.

These findings do not prove substantive research-policy misalignment. They show that shared SDG labels are insufficient evidence of shared semantic framing, and that embedding-based distance is a transparent diagnostic map of proximity and divergence. The results need caution because of register effects, policy source-family heterogeneity, OpenAlex retrieval boundaries, and an SDG 4 assignment that is artefact-affected rather than a clean education finding.

1 Introduction

Artificial intelligence is now routinely presented as part of the solution to sustainable development. In academic research, papers increasingly claim that a model, application, or dataset contributes to one or more Sustainable Development Goals (SDGs), with SDG attribution often functioning as a form of relevance-signalling beyond technical novelty. In international institutional SDG policy discourse, international organisations, intergovernmental bodies, and national AI strategies likewise frame AI as an enabler or accelerator of SDG progress (OECD, 2019; UK Government, 2021; UN AI Advisory Body, 2024; UNESCO, 2021; United Nations, 2015). Across these domains, a shared narrative has taken hold: AI is expected to help deliver the global development agenda.

That narrative matters because it shapes research framing, funding priorities, and policy expectations (Bush, 1945; Jasanoff, 2004; Stokes, 1997). Yet the apparent alignment it implies is usually inferred rather than demonstrated. Researchers and policymakers may converge on the same SDG labels while emphasising different problems, actors, timescales, and interventions. A paper and a policy document can both invoke SDG 3 (Good Health) or SDG 13 (Climate Action) without addressing the same substantive terrain. High-level thematic agreement can therefore mask deeper divergence. If so, current claims about "AI for SDGs" may overstate how far research effort is aligned with policy priorities, not because the narrative is necessarily false, but because the methods used to validate it remain too coarse.

The underlying problem is not new. Questions of research-policy divergence have been a central concern in policy studies and knowledge-utilisation scholarship for decades (Caplan, 1979; Guston, 2001; Haas, 1992; Weiss, 1979). What is new is not the embedding methodology itself, but its application to direct research–policy comparison within a shared SDG space. These tools allow broad discourse-level proximity to be assessed as an empirical property of text, rather than inferred only from anecdote, keyword coincidence, citation or mention counts, or aggregate coverage measures. In that sense, the aim of this paper is to map how closely academic AI-for-sustainability research and international institutional SDG policy discourse occupy nearby regions in semantic space.

Bibliometric work shows a persistent research-attention pattern, but rarely tests whether shared SDG labels imply shared semantic framing between research and policy discourse (reviewed in Section 2).

This study addresses that gap. Its central claim is straightforward: shared SDG labels are insufficient evidence of shared framing. I make three contributions. First, I advance AI–SDG mapping from Level 1 (lexical or bibliometric correspondence) to Level 2 (observed textual-semantic proximity in embedding space; Section 3.1). I treat this proximity as a register-sensitive proxy for framing, not as evidence of substantive alignment, using

Sentence-BERT (Reimers & Gurevych, 2019) embeddings and SDG centroids from four independent annotated corpora. Second, I distinguish between a *coverage gap* (which SDGs each corpus emphasises) and a *semantic gap* (how differently the two corpora discuss the same SDG), and measure both across all 17 SDGs. Third, I test whether coverage and framing are related across the 17 SDGs, or whether SDGs that both corpora cover can still be discussed in very different ways.

The central research question is therefore: *To what extent do academic AI-for-sustainability research and international institutional SDG policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?*

In brief, the two corpora differ substantially in both what they prioritise and how they frame those priorities, and the pattern of divergence is informative.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature on AI-SDG mapping, research-policy alignment, and semantic similarity methods. Section 3 describes the corpora, measurement instrument, and gap analysis procedures. Section 4 presents the empirical findings. Section 5 interprets the findings and discusses limitations. Section 6 concludes with implications and directions for future research.

2 Literature Review

2.1 AI and the SDGs: What Bibliometrics Shows

This section reviews what past studies say about AI and the SDGs. Most ask which goals draw the most research attention.

The best-known work, such as Vinuesa et al. (2020)'s assessment of AI's effects on the 169 SDG targets, rests on expert views or counts rather than on the texts themselves.

Broader SDG scientometric work has mapped the growth, institutional distribution, thematic structure, and interlinkages of MDG/SDG-related research across methods and time windows (Bautista-Puig et al., 2021; Cowls et al., 2021; Nedungadi et al., 2024; Singh et al., 2024). Bautista-Puig et al. (2021) chart the general SDG research landscape across higher-education and research institutions from 2000 to 2017 using a citation-based corpus of roughly 25,000 publications, finding SDG 3 (Good Health) by far the most addressed goal and SDG 7 and SDG 12 among the least. The AI-specific inventories converge on SDG 3 as dominant: Singh et al. (2024) map 20,511 AI publications (2001–2020) and rank SDG 3 and SDG 7 as the most applications-rich goals with SDG 17 the sparsest, while Cowls et al. (2021) survey 108 AI-for-social-good projects and report SDG 3 as the leader and SDGs 5, 16, and 17 each addressed by fewer than five. Nedungadi et al. (2024) apply BERTopic modelling to 1,288 big-data and AI publications (2013–2023) and likewise find SDG 3 most prominent, with SDG 9 and SDG 7 next. Across these studies, SDG 3 is the one robustly dominant goal, while the least-covered goals differ between the AI-specific inventories and the broader bibliometric picture — but none of them measures within-SDG semantic framing.

The methodological concern raised by Armitage et al. (2020) is important: SDG publication mapping depends on interpretive choices about what counts as SDG relevance, how target concepts are translated into queries or labels, and which bibliographic database is used. The embedding-based approach used here reduces some vocabulary-level brittleness of keyword dictionaries, but it remains a distributional measurement strategy whose results are conditional on corpus construction, centroid provenance, and model choice. Later benchmarking work reaches the same caution from a broader comparison of Boolean, machine-learning, and hybrid SDG-mapping systems: Kashnitsky et al. (2024) find that no single approach performs best across all validation datasets, and that no single "golden" SDG validation set exists.

Not only are SDG mapping systems methodologically contingent; the SDGs themselves are interdependent by design. Nilsson et al. (2016) observe that "the goals depend on each other" but that no operational framework specifies how, warning that treating targets in isolation "risk[s] perverse outcomes." Single-SDG assignment is therefore a simplifying assumption, not a reflection of how goals operate in practice.

One SDG breaks the interdependence pattern. Le Blanc (2015), in a network analysis of the proposed SDGs, treats SDG 17 (Partnerships for the Goals) as the dedicated goal for cross-cutting means of implementation — finance, technology transfer, capacity-building, and partnerships — and excludes it from cross-goal network mapping because its targets are implementation infrastructure rather than substantive domains: a distinction unique among the 17 goals. The Griggs et al. (2017) guide likewise describes SDG 17 as the main enabler of the whole framework. Unlike every other SDG, whose discourse centres on a shared substantive topic, SDG 17 policy discourse concerns coordinating and resourcing implementation itself — a conceptual space that technical AI research abstracts are not built to occupy.

Past work shows which SDGs attract AI and sustainability research, and how much that mapping depends on method and corpus. Whether research and policy texts for the same goals actually align is the subject of the next section.

2.2 Research-Policy Alignment in AI and Sustainability

This section asks whether AI research and sustainability policy line up, and why past work says they often do not.

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires. Strauss et al. (2025) show that corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the real-world deployment impacts that governance and policy must address. Sioumalas-Christodoulou and Tympas (2025) find that global AI metrics focus on technical and economic performance while policy discourse emphasises ethical and social dimensions aligned with SDG-related concerns. Toney-Wails et al. (2024), in the trustworthy-AI governance literature, provide a close methodological precedent, showing that policy and research communities can use shared trustworthy-AI terminology while differing in term salience, definitions, and focal concerns.

McCarthy et al. (2026) report a parallel finding in conservation science, where a multi-target SciBERT classifier applied to Ross Sea MPA research reveals that 59% of policy-prioritised research topics appear in fewer than 15 papers, demonstrating that research-policy coverage gaps are observable across domains.

Adjacent scientometric work has also used policy documents to measure research-policy connection, but at the level of mentions rather than semantic framing. Bornmann et al. (2016) examine climate-change publications in Altmetric-tracked policy documents and find that only 1.2% of 191,276 publications had at least one policy mention. Their study is useful here as a boundary marker: policy-document mentions can indicate one narrow form of policy uptake, but they do not show how research is discussed, interpreted, or framed inside policy discourse. The present study therefore addresses a different layer of the research-policy relationship: not whether policy documents cite particular papers, but whether research and policy-facing texts assigned to the same SDG occupy nearby regions in semantic space.

A related research-evaluation literature treats the research-policy relationship through the lens of societal impact measurement. Bornmann (2017) argues that measuring societal impact is harder than measuring scientific impact because the target of impact must first be specified: research may affect policymakers, business, culture, public health, or other social domains. Policy documents are therefore useful as one possible target-oriented source for measuring impact on policymaking, but this is a different task from semantic framing comparison. The present study does not ask whether research has achieved policy impact; it asks whether research and policy-facing texts occupy similar semantic regions when organised through shared SDG labels.

These findings are consistent with theoretical accounts of research-policy misalignment. Heink et al. (2015) identify credibility, relevance, and legitimacy as determinants of whether scientific knowledge informs policy; topical relevance alone is insufficient — relevance also depends on whether research results can shape actual decisions. In sustainability science, Cash et al. (2003) make a closely related argument: knowledge systems are more likely to support action when they produce information that is simultaneously salient, credible, and legitimate, and when they manage the boundary between knowledge and action through communication, translation, and mediation. The epistemic communities framework (Haas, 1992) provides a theoretical reason why shared framing matters for policy coordination under uncertainty: expert communities can influence policy by articulating cause–effect relationships, framing issues for collective debate, proposing policies, and identifying salient points for negotiation. However, Haas's concept is broader than semantic proximity alone, involving shared normative beliefs, causal beliefs, validity criteria, and a common policy enterprise. The present study therefore does not operationalise an epistemic community directly; it measures the weaker condition of observed semantic proximity between research and policy-facing discourse.

Cross (2013) strengthens this caution by arguing that epistemic communities should not be treated as simply present or absent: their influence varies with internal cohesion, professionalism, access to decision-makers, competition with other actors, and broader political context. This means that semantic proximity can at most indicate a possible precondition for knowledge transfer, not the existence, strength, or influence of an epistemic community.

The SDG framework is itself a politically negotiated instrument with documented limitations. Fukuda-Parr (2016) argues that global goals can have "distorting effects, redefining the meaning of development, and shaping policy by creating incentives to over-focus on target achievement." Bexell and Jönsson (2017) find that SDG summit documents remain "state-centric with great room for state sovereignty," avoid assigning causal responsibility for structural problems, and rely on "weak and unsystematic accountability."

A more fundamental tension exists within the framework itself. Hickel (2019) demonstrates that "Goal 8 violates the sustainability objectives of the SDGs" — 3% annual global growth is empirically incompatible with the resource-use and emissions targets in Goals 6, 12, 13, 14, and 15. The SDGs "lack theoretical grounding" and embody a contradiction between growth and ecological sustainability.

Heilinger et al. (2024) warns that the "sustainable AI" label can be used for greenwashing where technologies are framed as contributing to the SDGs without sufficient basis. Heras-Saizarbitoria et al. (2022), analysing 1,370 sustainability reports, find that most organisations' SDG engagement is superficial — "the SDGs only serve to add colour and fancy icons to reports" — suggesting that SDG-washing is a measurable empirical pattern. This supports the caution that reported alignment should not be conflated with substantive commitment.

Measuring semantic alignment with SDG centroids therefore uses a benchmark that is itself politically negotiated and internally inconsistent — not a neutral standard. Gibbons et al. (1994) distinguish Mode 1 knowledge production, where problems are set and solved within academic disciplinary interests, from Mode 2, where problems arise in contexts of application; to the extent that research follows Mode 1, its agendas may reflect internal disciplinary dynamics rather than external policy needs. This observation also operates alongside a further ambiguity: what it means to be "aligned" with a framework whose component goals are in tension with each other, and it suggests that divergence is not uniformly problematic — where research explores possibilities that policy discourse has not yet absorbed, distance may reflect productive exploration rather than failure of alignment.

The two-communities theory (Caplan, 1979) holds that researchers and policy makers live in separate worlds with different values, reward systems, and languages, so divergence is the default state of research-policy interaction rather than a failure. Taken together, this literature indicates that the divergence between AI research and sustainability policy is a substantive feature of the two knowledge systems rather than an artefact of measurement. Existing studies document the gap through coverage, citation, and impact-based methods, while theory explains it through legitimacy, expert communities, and the structural divide between academic and policy communities. None of these studies measures semantic framing within a single SDG. This study measures how close research and policy texts lie in semantic space; that closeness is a first step, not proof of alignment.

2.3 Semantic Methods: Rationale and Cautions

This section explains the method behind this study and why it needs care.

The use of Sentence-BERT (Reimers & Gurevych, 2019) and related transformer-based encoders for semantic comparison has been validated across several domains relevant to this study, but applying them to compare two distinct discourse corpora — research abstracts and policy documents — is a novel usage rather than a directly benchmarked one. This method follows the distributional hypothesis that meaning can be described through patterns of linguistic co-occurrence, a principle whose modern computational implementations trace back to Harris (1954)'s foundational observation that the distribution of a linguistic element is "the sum of all its environments." Sentence-BERT and related transformer-based encoders are therefore used here as pragmatic tools for large-scale semantic comparison, while the interpretation of distance remains descriptive rather than causal. Gjorgjevikj et al. (2025) benchmarked sentence encoders specifically for SDG association tasks, finding that general-purpose models from the sentence-transformers family perform competitively as baselines, and that domain-specific fine-tuning further improves predictive performance while reducing sensitivity to description length.

Rodriguez et al. (2023)'s conText framework provides a closer computational-social-science precedent for comparing meaning across contexts. Their embedding-regression approach estimates how focal terms are used differently across covariates such as party, time, or corpus, while explicitly treating embedding-based "meaning" as a thin distributional description rather than a causal model of human interpretation. This study does not adopt their word-level regression estimator, but it follows the same broad premise that semantic representations can be used descriptively to compare how different discourse communities frame shared terms or categories.

McCarthy et al. (2026) supply a yet closer empirical precedent for the present design. Applying a multi-target SciBERT classifier to Ross Sea MPA research and policy-prioritised topics, they find that 59% of policy-prioritised research topics appear in fewer than 15 papers, showing that research–policy coverage gaps are measurable with transformer encoders outside the AI–SDG setting. This study extends that logic from a single conservation domain to the full SDG landscape, and from coverage to within-SDG semantic framing.

An older corpus-linguistic tradition gives a further reason for caution when interpreting cross-corpus semantic distance. Biber's multidimensional analysis of spoken and written English showed that register differences are not simple binary contrasts, but continuous patterns of co-occurring linguistic features across genres (Biber, 1988). For the present study, this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function. Embedding distance is therefore treated as a measure of observed textual-semantic proximity, not as a purified measure of topic alone.

Embedding models let us compare research and policy texts, but applying them across two corpora is new. They represent meaning through distribution, not cause, and cannot separate topic from register. The measures used here are descriptive: they report how close or distant the two corpora are in semantic space, not whether that closeness is real alignment.

2.4 Research Gap

To the best of my knowledge, the literature reviewed above has not produced: (i) a systematic comparison of how academic AI-for-sustainability research and institutional SDG policy discourse differ in SDG coverage and within-SDG semantic framing — prior work measures research coverage, citation patterns, or policy-document uptake, but does not compare the two corpora directly on these dimensions; or (ii) an examination of whether coverage gaps and semantic gaps are positively related, weakly related, or inversely related. These are the contributions of the present study, and they answer the central research question set out in the Introduction. Together, they provide a systematic framework for distinguishing research-policy coverage from research-policy semantic proximity within a shared SDG representation. This framework is deliberately diagnostic: it maps where observed divergence is visible, without classifying whether any given gap reflects productive or problematic disconnection — the latter would require qualitative follow-up beyond the scope of embedding analysis.

3 Methodology

3.1 A Three-Level Framework of Research-Policy Alignment

This study measures *topical overlap* between research and policy-facing discourse using semantic proximity in embedding space. I do not claim to measure substantive alignment, causal influence, or operational alignment — shared vocabulary does not guarantee shared meaning, but its absence makes meaningful dialogue structurally impossible. Nor should the measure be interpreted as a societal-impact metric: it captures observed textual-semantic distance between two corpora, not policy uptake (Bornmann et al., 2016) or shared causal beliefs (Haas, 1992). Embedding distances describe differences in textual context and usage, not direct effects on human understanding or policy uptake.

Following Biber and Conrad's distinction between register, genre, and style (Biber & Conrad, 2009), I treat the research-policy contrast primarily as a register contrast — the two corpora are produced in different institutional settings for different audiences and purposes. Cross-corpus register differences are discussed in Section 2.3 and bounded in Section 5.4.

I propose that research-policy alignment be understood as a three-level construct rather than a binary property of shared labels:

Level 1 — lexical or bibliometric correspondence. The dominant mode of existing SDG-mapping work, assigning texts to goals through keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). Necessary but weak: shared vocabulary does not guarantee shared meaning (Bornmann et al., 2016), and keyword methods are brittle to vocabulary and database choices (Armitage et al., 2020).

Level 2 — observed textual-semantic proximity in embedding space. Sentence embeddings locate texts in a shared semantic space where proximity reflects distributional patterns of co-occurrence (Harris, 1954; Reimers & Gurevych, 2019), and encoder-based SDG association has been benchmarked as a competitive baseline (Gjorgjevikj et al., 2025). This level detects divergence that keyword methods miss, while remaining a proxy rather than a verdict: embedding distance mixes modality, abstraction, institutional stance, and theme (Biber, 1988), and the research-policy contrast is primarily a register contrast (Biber & Conrad, 2009). The present study is positioned at this level.

Level 3 — substantive comparison of implied problems, actors, and interventions. The ultimate object of alignment is whether the two discourses address the same problems, involve the same actors, and propose the same interventions (Cash et al., 2003; Guston, 2001; Haas, 1992). Because this requires interpreting causal beliefs, legitimacy, and institutional mechanism (Caplan, 1979; Haas, 1992), it lies beyond unsupervised text analysis and is left to qualitative follow-up.

Advancing from Level 1 to Level 2 is the study's concrete methodological contribution; Level 3 names the boundary it deliberately does not cross.

3.2 Research Corpus

This study uses only publicly available text data and codebase. It does not involve human participants or personal data. No ethical approval was required.

The research corpus consists of 2,543,698 English-language research abstracts retrieved from the OpenAlex API (Priem et al., 2022), covering publications between 2018 and 2025. Abstracts were retrieved by combining OpenAlex's native SDG work filter with four AI terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`) — 68 queries (17 SDGs × 4 terms). Complete query parameters are documented in `code/0_fetch/fetch_openalex.py`. OpenAlex SDG metadata defines the candidate universe; the SDG assignments used in the analysis are produced by the independently validated centroid instrument (Section 3.5).

The choice of the four AI retrieval terms — `artificial intelligence`, `machine learning`, `deep learning`, `neural network` — was made on initial intuition rather than any systematic or validated basis, and was fixed before analysis began; it is therefore not defended or varied here. The corpus should be understood as conditional on this term selection, which may under- or over-represent subfields of AI-for-Sustainability research. The 2018 start date reflects the post-AlexNet (Krizhevsky et al., 2012) emergence of deep learning as the dominant paradigm and a reasonable research-uptake lag after the SDGs' adoption in 2015.

An important asymmetry follows: the research corpus is a tight intersection of AI and SDG content (AI ∩ SDG), while the policy corpus (Section 3.3) is a broader union covering institutional SDG discourse and AI governance. Between-SDG rankings should be interpreted as conditional on this asymmetric construction. Following Armitage et al. (2020), this retrieval universe is a method-dependent corpus boundary, not a neutral population of all SDG-relevant AI research. After deduplication and removal of records lacking usable abstracts, 2,543,698 unique papers were retained. No balancing by SDG or citation count is applied — trimming would distort the coverage profile the study aims to measure.

3.3 Policy Corpus

The policy corpus contains 54,082 text segments from three collections, representing 2,456 unique source documents. It should be read as mixed *international institutional SDG policy discourse* rather than the full universe of AI sustainability policy:

1. **Curated AI and SDG policy documents** (15,639 segments, 95 documents): UN, OECD, EU, UK, and AU AI governance frameworks; IPCC assessment reports; SDSN sustainable development reports; and national AI strategies.
2. **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024) (31,971 segments, 313 documents): National and local self-assessments submitted to the UN High-Level Political Forum.
3. **UN General Debate Corpus** (Harvard Dataverse, 2024) (6,472 segments, 2,048 documents): SDG-relevant passages from General Debate speeches, sessions 70–80 (2015–2024).

Each source document was segmented into 150–300 word units using sentence-boundary-aware dynamic segmentation; fragments under 20 words were discarded. After deduplication by exact text match, 54,082 unique segments remained. Segment identifiers encode source document identity for document-level weighting.

Source heterogeneity note. The three collections differ in genre and scope — the curated subset is AI-governance-focused, while SDGi and UNGDC reflect broader institutional SDG discourse. The full-corpus policy distribution should be read accordingly; the curated sub-corpus (95 documents) is the more relevant reference for AI-governance-specific interpretation. Appendix A.2 reports a source-family sensitivity check.

3.4 Embedding Model and Normalisation

All texts were encoded using `all-MiniLM-L6-v2` (Sentence-Transformers, 2021a) from the sentence-transformers library (Reimers & Gurevych, 2019), producing 384-dimensional dense vector representations. This pre-trained model achieves competitive performance on semantic textual similarity benchmarks, runs efficiently on CPU hardware, and requires no fine-tuning on the data under analysis — the embedding space is a stable pre-existing coordinate system. All embeddings were L2-normalised to unit vectors so that cosine similarity between any two vectors equals their dot product.

The canonical analysis uses MiniLM; all key findings are replicated using `all-mpnet-base-v2`

(768-dimensional) (Sentence-Transformers, 2021b) as an alternative encoder, with results reported in the cross-sensitivity comparison (Table 3) and Appendix D.

3.5 SDG Measurement Instrument: Centroids and Validation

A centroid is the mean embedding of all labelled texts for a given SDG, representing that goal's typical semantic location. The instrument comprises 17 per-SDG centroid vectors built from four independent annotated corpora: the OSDG Community Dataset (Pukelis et al., 2022) (30,534 texts, SDGs 1–16), the IISD SDG Knowledge Hub (Wulff et al., 2023) (2,221 texts, SDGs 1–17), the SDGi Corpus (SDGi, 2024) (5,233 texts, SDGs 1–17), and the Aurora dataset (Vanderfeesten et al., 2020) (5,619 expert-validated texts, SDGs 1–17). For SDG 17, which OSDG does not cover, the centroid draws on 616 Knowledge Hub, 321 SDGi, and 557 Aurora texts. The SDG Classification Benchmark (SDG Classification Expert Group, 2025) is held out as an independent validation set. Each centroid is L2-normalised so that downstream dot products equal cosine similarities.

All four corpora supply human- or expert-validated labels rather than machine-generated ones, and they differ in provenance. OSDG and SDGi are policy-adjacent: OSDG volunteers confirm a suggested label across multiple annotators, and SDGi compiles government Voluntary National and Local Reviews submitted to the UN. The SDG Knowledge Hub is journalism and reporting; Aurora is academic, expert-validated text. The mix broadens the label base, but OSDG alone contributes about 70% of all labelled texts and draws on UN-adjacent documents, so the centroids remain weighted toward policy vocabulary — a potential bias the analysis tests for explicitly. No high-quality graded relevance sets exist for SDG classification, making this combined, multi-source instrument the best available resource (Kashnitsky et al., 2024).

A text is assigned to the nearest SDG centroid under cosine similarity. Every assignment is traceable to a single vector comparison — the embedding space is a frozen coordinate system, not an optimised classifier. The instrument achieves macro-F1 = 0.716 on the held-out benchmark (616 texts, all 17 SDGs), above the pre-registered pass threshold of 0.50 and far above the random 17-class baseline of 0.059. This performance is not suitable for operational labelling but is sufficiently discriminative for exploratory corpus-level comparison. Macro-F1 is the unweighted mean of per-SDG scores; those scores are reported in Table 1 alongside SDG-specific concerns that inform the interpretation of the results below.

The validation task tests whether benchmark examples are assigned to the nearest appropriate SDG; it does not evaluate a rejection option for non-SDG texts. This is acceptable because the research corpus is first restricted to an OpenAlex AI-and-SDG universe — the centroid instrument then answers a relative question: among the 17 SDGs, which is this text closest to?

Nearest-centroid assigns each text to the SDG whose average embedding is closest. The method is simple, fast, and avoids the overfitting that small training sets bring to supervised classifiers. I chose it deliberately: the study needs a transparent tool to compare corpora, not a classifier built to label single documents. A tuned classifier could label single documents more accurately, but it would add training choices, hyperparameters, and overfitting risk I do not need. The centroid is the average point of each SDG's texts in semantic space, so the scores I report are easy to interpret.

Additional diagnostic checks — PCA visualisations, centroid-structure diagnostics, a softmax-weighted multi-label alternative, policy source-family sensitivity, and an SDG 4 lexical audit — are reported in Appendix A. The canonical analysis uses hard nearest-centroid assignment as a transparent and reproducible anchoring procedure.

3.6 Coverage Gap Analysis

For each corpus item, the predicted SDG is the nearest centroid under cosine similarity (hard assignment). The coverage profile is the proportion of items assigned to each SDG.

Because the policy source collections differ dramatically in size (SDGi: 31,971 segments; curated: 15,639; UNGDC: 6,472), a segment-weighted profile would be dominated by a small number of large documents. I therefore compute a *document-weighted* profile in which each of the 2,456 unique source documents contributes equally. Each document's SDG is the goal with the highest mean segment score. Document-weighted profiles are the canonical representation; segment-weighted profiles are retained as a diagnostic comparison.

The coverage gap for SDG j is defined as:

$$\text{CoverageGap}_j = \left|\text{ResearchProportion}_j - \text{PolicyProportion}_j^{\text{docwt}}\right|$$

The signed version of the same quantity is:

$$\text{DominanceSigned}_j = \text{ResearchProportion}_j - \text{PolicyProportion}_j^{\text{docwt}}$$

A positive value means research dominates SDG j; a negative value means policy dominates. These two definitions, together with research coverage and policy coverage, are the four predictors reported in Table 2.

3.7 Semantic Gap Analysis

The semantic gap measures within-SDG divergence: given that both corpora have texts assigned to SDG j, how differently do they discuss it? For each SDG j, I collect all assigned research papers and all assigned policy segments (capped at 50 per source document to prevent single-document dominance — SDSN contributes up to 3,179 segments otherwise), compute the L2-normalised mean embedding of each cluster ($\mathbf{c}_j^{\text{res}}$, $\mathbf{c}_j^{\text{pol}}$), and define:

$$\text{SemanticGap}_j = 1 - (\mathbf{c}_j^{\text{res}} \cdot \mathbf{c}_j^{\text{pol}})$$

A gap of 0 indicates perfectly overlapping semantic directions; a gap approaching 1 indicates orthogonal framing. Robustness was confirmed by replicating with caps of 20 and 100 segments per document — rankings were stable across all three.

The raw centroid distance is retained as the primary measure because it directly corresponds to the observed framing difference between the two corpora within the same SDG. Register-adjusted sensitivity checks (Appendix E) are reported as uncertainty bounds rather than replacement estimates — more aggressive procedures risk over-subtracting the substantive contrast of interest.

3.8 Coverage–Semantic Interaction

The first hypothesis (H1) decomposes into four named sub-hypotheses, each correlating a distinct coverage predictor with the within-SDG semantic gap across the 17 SDGs: H1a, coverage gap $\leftrightarrow$ semantic gap; H1b, policy–research dominance $\leftrightarrow$ semantic gap; H1c, research coverage $\leftrightarrow$ semantic gap; and H1d, policy coverage $\leftrightarrow$ semantic gap. H1a is the primary, motivating hypothesis: the absolute research–policy coverage gap is expected to predict semantic divergence. H1b is its directional complement, testing whether the signed balance of research and policy attention (which side dominates a goal) also predicts divergence. I compute Pearson r and Spearman ρ between each coverage predictor and semantic gap scores across all 17 SDGs, with sensitivity analyses excluding SDG 4 (see Section 3.9). Given $n = 17$, the tests have limited power to detect anything but a large linear effect; this is a scope condition.

3.9 Key Assumptions and Mitigations

Corpus-level score asymmetry. Policy segments score systematically higher than research papers against the combined centroids (mean 0.560 vs 0.232, difference 0.328). This may reflect genre effects from the multi-source centroid or the substantive fact that policy documents address SDG implementation more directly — the two cannot be separated. The ambiguity does not affect the primary findings: coverage uses hard assignment rather than absolute scores, and gap rankings are robust to the asymmetry. A leave-one-source-out check tests whether this asymmetry comes from SDGi texts feeding both the centroid and the scored policy corpus. It does not: see Appendix A.3.

Text-unit asymmetry. Research papers are single abstract-length texts (mean ∼200 words); policy items are 150-word segments from documents of highly variable length. Document-weighting and per-document segment caps are the primary mitigations.

SDG 4 lexical artefact. The research coverage assigns 14.0% of papers to SDG 4 (Quality Education), flagged as artefact-affected because ML papers share core vocabulary (“learning”, “training”, “model”) with the OSDG education corpus. Appendix A.4 confirms that SDG 4-assigned texts contain education-specific terms more often than matched non-SDG4 samples, but co-occur with ML-learning vocabulary. The result is retained as a learning-related SDG 4 assignment rather than a clean education estimate. Sensitivity analyses excluding SDG 4 are provided for all correlation results.

4 Results

This section reports what the instrument measured: the validation scores, the coverage and semantic gaps, the H1 correlation tests, and the robustness of those rankings. It describes the findings; it does not interpret them. Interpretation, limits, and implications follow in the Discussion (Section 5).

4.1 SDG Measurement Instrument Validation

Table 1 reports per-SDG F1 from the benchmark (all 17 SDGs, n = 616). The instrument reaches macro-F1 = 0.716, above the 0.50 pass threshold. Accuracy is 0.735; a 17-class random guess scores $1/17 \approx 0.059$, so the instrument performs 12.2 times better. I reached this score with no supervised training. It shows how far apart the 17 SDGs sit in the pre-trained space, not how well a classifier was tuned.

The scores show a limit of the method, not a flaw in the data. Kashnitsky et al. (2024) find that no SDG-mapping method performs best across validation sets, and Gjorgjevikj et al. (2025) show that off-the-shelf encoders such as mine have a known ceiling on fine-grained multi-label work. I read the benchmark as a map of where the 17 goals blur in semantic space, not as proof of precise per-SDG accuracy.

Some goals separate cleanly: SDG 4 (F1 = 0.907) and SDGs 6, 7, and 16 (each above 0.87) are assigned with confidence. Most goals cluster between F1 = 0.67 and 0.69, including SDG 15 and SDGs 1, 3, 9, 12, and 13. Bootstrap confidence intervals (Appendix 6) show this middle band cannot be distinguished from the macro-F1 mean or from each other. Only the high cluster — SDGs 4, 6, 7, 14, and 16 — and SDG 17 at the bottom are robustly

Table 1. Per-SDG F1 scores from benchmark validation across embedding models and reference sources (n = 616).

	Model		Reference source			
SDG	Can	MPNet	OSDG	SDGi	KH	Aur
SDG 1	0.687	0.676	0.697	0.708	0.000	0.522
SDG 2	0.854	0.848	0.844	0.780	0.792	0.567
SDG 3	0.680	0.630	0.583	0.607	0.678	0.630
SDG 4	0.907	0.918	0.920	0.879	0.857	0.872
SDG 5	0.737	0.667	0.712	0.658	0.649	0.638
SDG 6	0.878	0.857	0.869	0.839	0.848	0.650
SDG 7	0.878	0.885	0.871	0.869	0.860	0.882
SDG 8	0.548	0.554	0.523	0.407	0.362	0.327
SDG 9	0.678	0.727	0.618	0.615	0.552	0.160
SDG 10	0.733	0.655	0.712	0.600	0.150	0.108
SDG 11	0.545	0.549	0.509	0.571	0.630	0.538
SDG 12	0.684	0.725	0.701	0.699	0.659	0.508
SDG 13	0.694	0.693	0.644	0.623	0.604	0.227
SDG 14	0.822	0.811	0.822	0.789	0.784	0.613
SDG 15	0.683	0.753	0.683	0.699	0.667	0.571
SDG 16	0.870	0.853	0.845	0.800	0.780	0.781
SDG 17	0.300	0.302	–	0.437	0.316	0.267
Macro-F1 (SDGs 1–17)	**0.716**	**0.712**	**0.722**	**0.681**	**0.599**	**0.521**

Notes: Can = canonical (the combined MiniLM specification); MPNet = all-mpnet-base-v2; OSDG = OSDG Community Dataset; SDGi = SDGi Corpus; KH = Knowledge Hub; Aur = Aurora. The benchmark is fully independent of centroid construction. Macro-F1 is the unweighted mean over available SDGs per configuration (OSDG covers 16 SDGs; all others cover 17). Bootstrap 95% confidence intervals for the canonical column are reported in Appendix 6.

separated from the rest. This separation holds for the large majority of these scores across the reference sources used for centroid construction (Table 1). The centroid-similarity heatmap (Figure 4) and the full confusion matrix (Table 10) in Appendix A.5 show these proximities and the per-SDG errors directly.

SDG 17 (Partnerships for the Goals) has the lowest F1 of the 17 goals (0.300). Le Blanc (2015) and Griggs et al. (2017) treat SDG 17 as the cross-cutting means-of-implementation goal and exclude it from single-goal mappings, because its mandate resists a compact definition; this is why its classification is the least certain, and per-source validation (Appendix A.1) confirms the weak score holds across all four centroid sources.

4.2 Coverage Gap: What SDGs Each Corpus Prioritises

Figure 1 presents the document-weighted policy coverage profile alongside the research coverage profile. The two profiles are markedly different.

Policy corpus. Three SDGs dominate the document-weighted profile of the full institutional policy corpus: SDG 13 (Climate Action, 25.0%), SDG 17 (Partnerships for the Goals, 48.2%), and SDG 16 (Peace, Justice and Strong Institutions, 14.5%), accounting for 87.7% of policy documents. The SDG 17 share is highly sensitive to reference-source

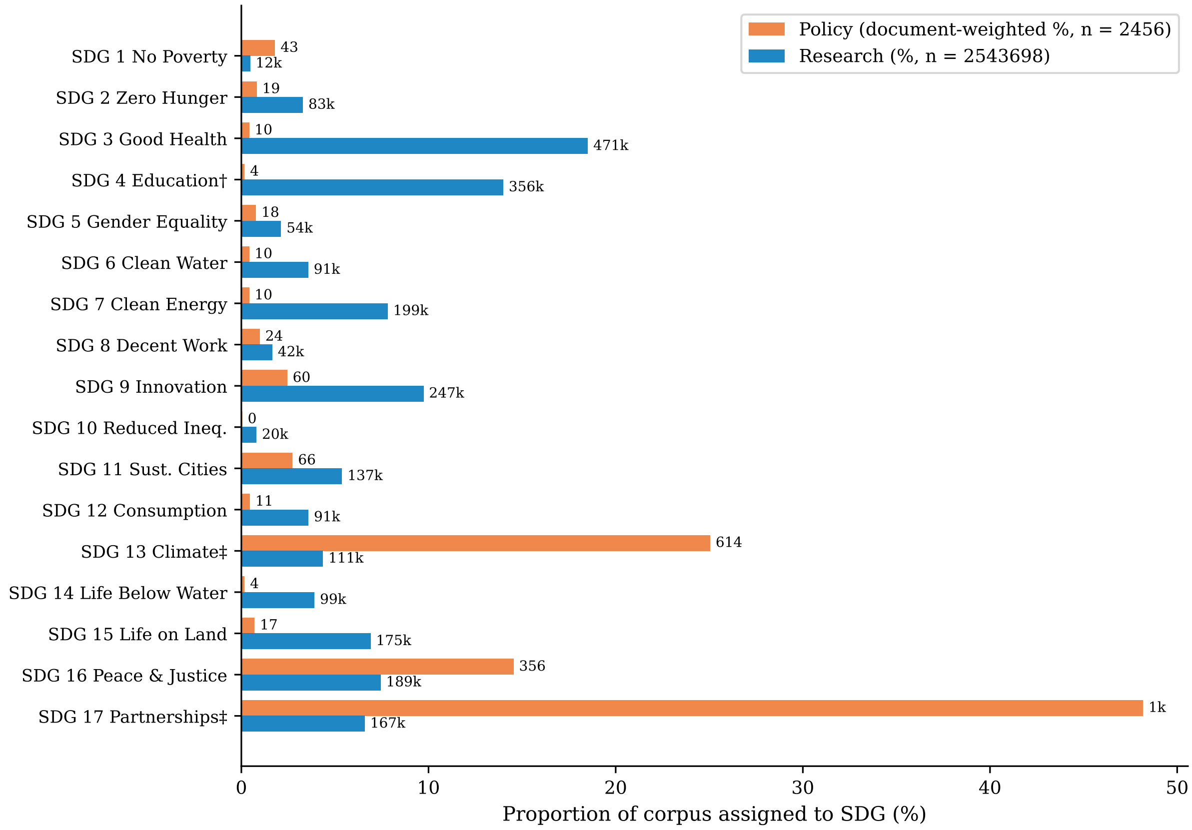

Figure 1. Coverage profiles by SDG: proportion of each corpus assigned to each SDG via hard assignment (highest-scoring SDG). Policy proportions are document-weighted (n = 2,456 source documents); research proportions are paper-weighted (n = 2,543,698 papers).

Notes: [(a)]SDG 4 research share likely inflated by ML vocabulary artefact (see Section 3.9); its F1 = 0.907 in Table 1 indicates confident assignment, which strengthens the artefact concern. Total coverage gap across all SDGs = 1.414 (mean per-SDG = 0.083).

choice (Section 5.4). These three SDGs form a governance-and-climate cluster distinct from sector-specific programme areas. Appendix A.2 shows that this full-corpus profile is source-family sensitive: the curated AI/SDG sub-corpus is much more innovation-heavy, while the broader SDGi and UNGDC components are more climate-, partnerships-, and governance-oriented.

Research corpus. The research profile is broader, but it still concentrates strongly on a smaller set of SDGs. The largest research shares are SDG 3 (Good Health, 18.5%), SDG 4 (Quality Education; flagged as artefact-affected, 14.0%), and SDG 9 (Innovation, 9.7%), followed by SDG 16 and SDG 7 at 7.4% and 7.8% respectively. Appendix A.4 reports a lexical audit supporting the SDG 4 concern. Even with that caution, the research corpus clearly does not mirror the policy-side dominance of SDGs 13, 17, and 16.

Largest coverage mismatches. The five largest absolute coverage gaps are SDG 17 (0.416), SDG 13 (0.207), SDG 3 (0.181), SDG 4 (0.138), and SDG 7 (0.074).

Score note. The mean top-centroid score for policy segments (0.560) exceeds that for research papers (0.232) by 0.328, above the pre-registered flag threshold of 0.10. Coverage proportions are based on hard assignment (which SDG scores highest) rather than absolute score levels, so the ambiguity does not affect the primary findings.

4.3 Semantic Gap: How Differently Research and Policy Discuss Each SDG

Figure 2 shows the within-SDG semantic gap for all 17 SDGs. Semantic gaps range from 0.234 (SDG 9) to 0.671 (SDG 17), a span of 0.437 cosine distance units. The SDG 17 gap is the largest but is highly sensitive to reference-source choice (Section 5.4). For context, the largest gap (0.671) is roughly double the systematic policy–research score gap (0.328, Section 3.9), indicating that the within-SDG divergence substantially exceeds the baseline measurement asymmetry. Rankings are stable across segment caps of 20, 50, and 100, confirming robustness to the document sampling parameter.

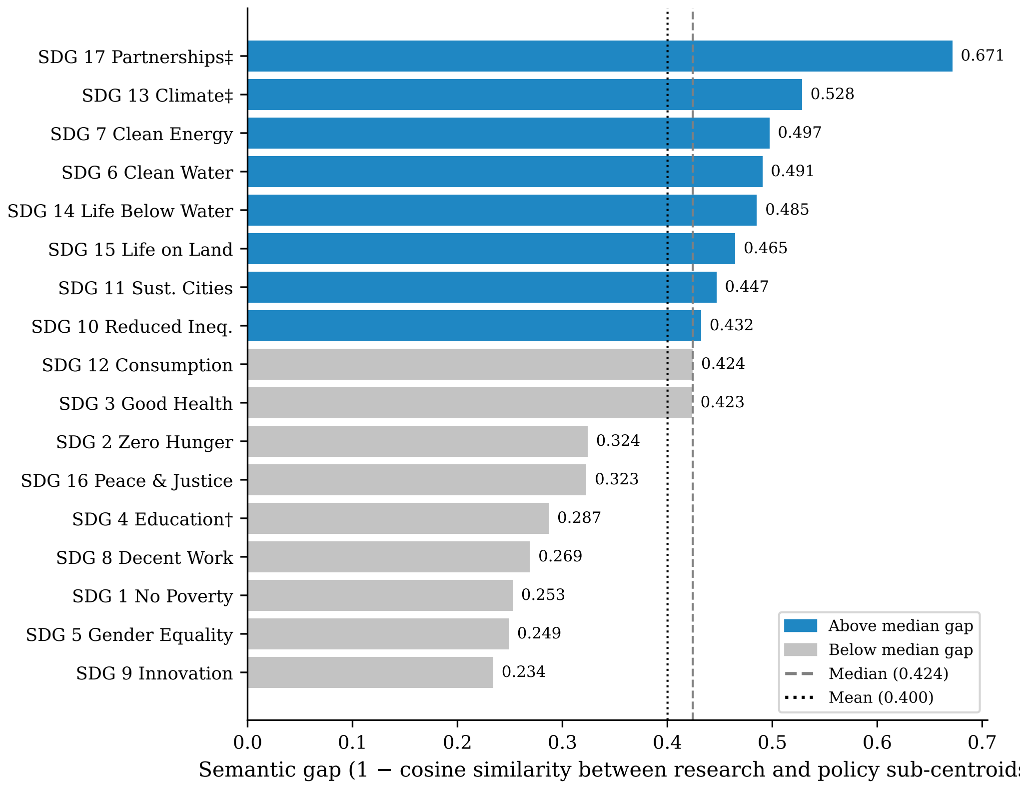

Figure 2. Within-SDG semantic gap by SDG, sorted descending. Semantic gap = 1− cosine similarity between the research sub-centroid and the policy sub-centroid for each SDG (segment cap = 50 per source document). Dashed line = median (0.424); dotted line = mean (0.400). Dark bars exceed the median; light bars fall below it.

Largest semantic gaps. The greatest within-SDG divergence appears in SDG 17 (Partnerships, 0.671), SDG 13 (Climate Action, 0.528), and SDG 10 (Reduced Inequalities, 0.432), with SDGs 6 and 7 close behind. This pattern is robust across reference sources: single-source centroids produce similar validation F1 scores to the combined centroid for most SDGs, with OSDG-only and combined F1 agreeing within 0.05 for most of the 16 available SDGs (Appendix A.1). Notable exceptions are SDG 10 (Reduced Inequalities), where the Knowledge Hub centroid produces F1 = 0.150 (vs combined 0.733), and SDG 8 (Decent Work), where both SDGi-only and KH-only centroids underperform the combined (0.407 and 0.362 vs 0.548). The combined centroid now also incorporates the Aurora expert-validated research corpus (Vanderfeesten et al., 2020) (5,619 texts, all 17 SDGs), adding a research-domain perspective alongside the OSDG, SDGi, and Knowledge Hub sources. Appendix A.1 reports per-source validation for all five sources. Given the centroid collinearity noted in Section 4.2, gap estimates for SDGs 13 and 17 individually should be read with the same caution; the cross-sensitivity check in Section 4.5 confirms that both show substantial rank variation under alternative reference sources.

Smallest semantic gaps. The lowest gaps occur in SDG 9 (0.234), SDG 5 (0.249), SDG 4 (0.287), and SDG 8 (0.269). Low semantic gap does not imply balanced attention: SDG 9 is still strongly research-dominant in coverage, and SDG 4 remains subject to the learning-vocabulary concern discussed in Appendix A.4. The key point is narrower: some SDGs show relatively close within-goal framing even when the two corpora differ sharply in how much attention they allocate to that goal. Appendix B.3 gives a compact lexical interpretability check for SDGs 17, 13, and 9 to show what these high- and lower-gap cases look like in text.

4.4 Coverage–Semantic Interaction: Two Distinct Dimensions of Observed Divergence

Table 2 reports the H1a–H1d correlation tests between coverage measures and semantic gap scores across the SDGs.

The four predictors are research coverage, policy coverage, the absolute research–policy coverage gap, and the signed policy–research dominance. Each panel reports the canonical result across all 17 observed SDGs and a sensitivity test excluding SDG 4, then replicates the four tests under an alternative embedding model (MPNet) and four alternative reference-source centroids (OSDG, SDGi, Knowledge Hub, Aurora).

Table 2. H1a–H1d correlation tests between coverage predictors and the within-SDG semantic gap.

Config	Pearson r	p	Spearman ρ	p
H1c. Research coverage ↔ Semantic gap				
Primary observed SDGs (n=17)	0.040	0.879	0.137	0.599
Excluding SDG 4 (n=16)	0.167	0.536	0.224	0.405
MPNet model (n=17)	-0.084	0.749	0.049	0.852
OSDG-only centroids (n=16)	-0.030	0.913	0.047	0.863
SDGi-only centroids (n=17)	0.122	0.642	0.176	0.498
Knowledge Hub-only centroids (n=17)	0.023	0.929	0.164	0.529
Aurora-only centroids (n=17)	-0.057	0.829	0.127	0.626
H1d. Policy coverage ↔ Semantic gap				
Primary observed SDGs (n=17)	0.582	0.014	-0.020	0.940
Excluding SDG 4 (n=16)	0.575	0.020	-0.077	0.778
MPNet model (n=17)	0.346	0.173	0.029	0.911
OSDG-only centroids (n=16)	0.060	0.824	-0.265	0.322
SDGi-only centroids (n=17)	-0.018	0.947	-0.206	0.428
Knowledge Hub-only centroids (n=17)	-0.617	0.008	-0.475	0.054
Aurora-only centroids (n=17)	-0.613	0.009	-0.792	0.000
H1a. Coverage gap ↔ Semantic gap				
Primary observed SDGs (n=17)	0.604	0.010	0.453	0.068
Excluding SDG 4 (n=16)	0.662	0.005	0.526	0.036
MPNet model (n=17)	0.383	0.129	0.316	0.216
OSDG-only centroids (n=16)	0.042	0.877	-0.279	0.295
SDGi-only centroids (n=17)	0.035	0.894	-0.096	0.715
Knowledge Hub-only centroids (n=17)	-0.247	0.339	-0.211	0.417
Aurora-only centroids (n=17)	-0.584	0.014	-0.554	0.021
H1b. Policy–research dominance ↔ Semantic gap				
Primary observed SDGs (n=17)	-0.534	0.027	-0.093	0.722
Excluding SDG 4 (n=16)	-0.503	0.047	-0.032	0.905
MPNet model (n=17)	-0.326	0.201	0.064	0.808
OSDG-only centroids (n=16)	-0.073	0.788	0.138	0.610
SDGi-only centroids (n=17)	0.128	0.624	0.240	0.353
Knowledge Hub-only centroids (n=17)	0.332	0.193	0.471	0.057
Aurora-only centroids (n=17)	0.401	0.111	0.331	0.195

The primary hypothesis, H1a, is supported in the canonical specification: the absolute research–policy coverage gap is positively related to semantic gap ($r = 0.604$, $p = 0.010$;

$n = 17$). H1c, the research-coverage predictor, is **not supported**: research coverage proportion and semantic gap are almost uncorrelated across SDGs ($r = 0.040$, $p = 0.879$; $n = 17$), and the 95% confidence interval for r spans approximately $[-0.45, +0.51]$, so the result provides low evidence for a linear relationship. Two further exploratory tests are reported: H1d, policy coverage alone, is positively related to divergence ($r = 0.582$, $p = 0.014$; $n = 17$); and H1b, the signed policy–research dominance, is negatively related ($r = -0.534$, $p = 0.027$) — SDGs where policy dominates the discourse diverge most. Both positive correlations reverse sign under alternative reference centroids: under Knowledge Hub-only centroids the policy-coverage correlation is $r = -0.617$ ($p = 0.008$) and the coverage-gap correlation is $r = -0.247$ ($p = 0.339$); under Aurora-only centroids they are $r = -0.613$ ($p = 0.009$) and $r = -0.584$ ($p = 0.014$) respectively (Table 2). This pattern is an exploratory result: across the four predictors and seven centroid-or-model configurations in Table 2 the study runs 28 correlation tests without a multiplicity correction, so the nominal p values of 0.010–0.014 should not be read as strong evidence on their own.

The diagnostic scatter plot (Figure 3) provides a per-SDG complement to the aggregate correlation tests. The formal coverage-gap test confirms the visual pattern: SDGs with larger absolute research–policy coverage gaps have larger semantic gaps, and the most policy-dominant SDGs (13, 17) sit above the median on both axes. The significance of the canonical coverage-gap association, however, does not survive the MPNet embedding (Table 2), so it should be read as a descriptive tendency rather than a robust effect.

Figure 3 illustrates this diagnostic refinement. High-coverage research goals are split between relatively low-gap cases such as SDG 9 and higher-gap cases such as SDG 3 and SDG 7, which is why the H1c research-coverage test is near zero. At the same time, the most policy-dominant goals, especially SDGs 13 and 17, sit above the median on both axes.

4.5 Cross-Sensitivity Robustness of the Gap Rankings

The semantic gap rankings discussed below are not equally stable across all measurement choices. Table 3 compares each SDG's gap rank under the canonical configuration (MiniLM encoder, combined centroids, full policy corpus) against three families of alternatives: an alternative embedding model (MPNet), five reference-source configurations (OSDG-only, SDGi-only, Knowledge Hub-only, Aurora-only), and four policy sub-corpora (curated AI/SDG, SDGi VNR/VLR, UNGDC). Each cell reports the within-SDG gap rank where 1 = largest gap.

Three tiers of robustness emerge directly from the data. First, a small set of SDGs maintain their relative positions across all or nearly all configurations. SDG 17 is the top-gap SDG in the canonical analysis and remains among the top two under model substitution and all

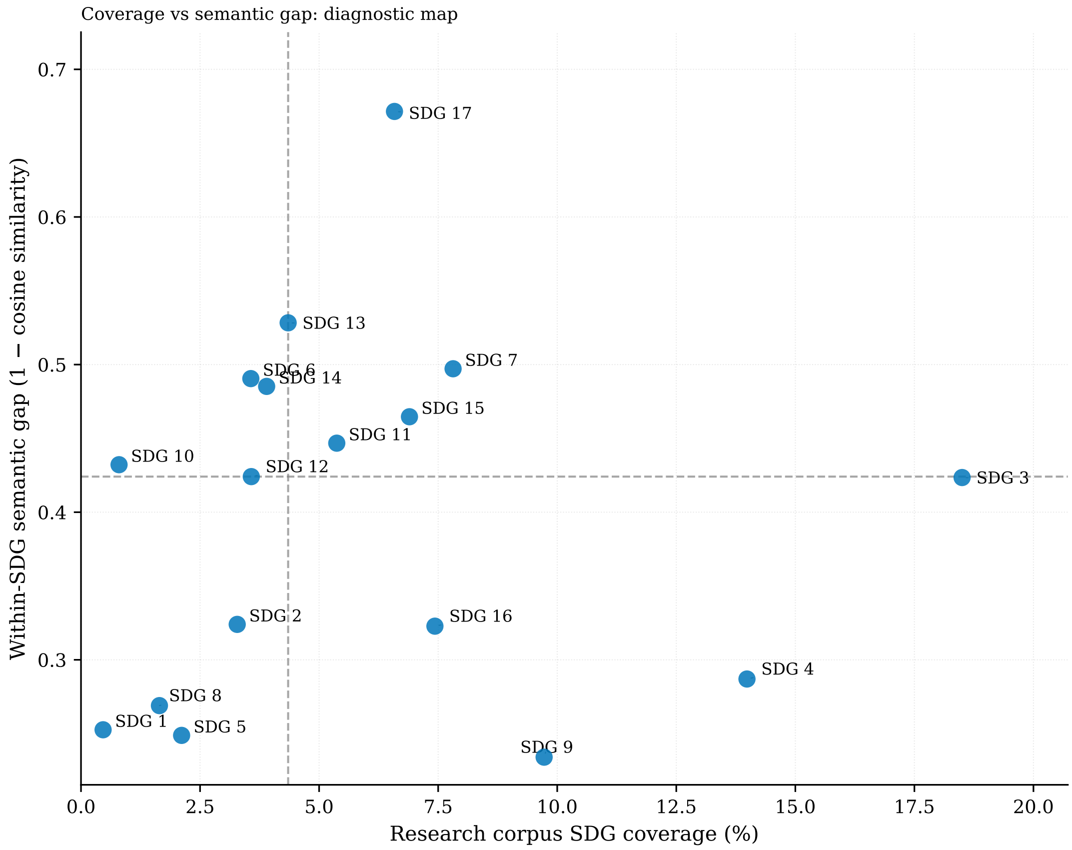

Figure 3. Coverage vs semantic gap scatter plot (diagnostic map). Each point is one SDG. Reference lines indicate the median research coverage (4.35%) and median semantic gap (0.424). The figure shows that high research attention can coexist with either low or high semantic divergence, while the most policy-dominant SDGs tend to sit above the median on both axes.

policy-family checks, though its rank varies considerably under alternative reference sources (rank 8 under SDGi-only centroids, rank 16–17 under Knowledge Hub and Aurora). SDGs 9 and 5 are consistently among the lowest-gap SDGs across all measurement configurations, typically occupying the bottom three ranks. Second, a cluster of SDGs (6, 7, 13, 14) consistently occupy the upper half of the gap ranking across all checks, though their exact ordering shifts by no more than three positions. Third, the remaining SDGs (1, 2, 3, 8, 10, 11, 12, 15, 16) are sensitive to measurement choices: their ranks shift by four or more positions under model substitution alone, with additional variation across reference and policy sources. The discussion below limits its interpretive claims to findings that survive this filtering.

Table 3. Cross-sensitivity robustness of within-SDG semantic gap rankings. Each cell reports the gap rank (1 = largest gap, 17 = smallest gap) under each measurement configuration. A "–" indicates that the source does not cover that SDG.

	Model		Reference source					Policy source			
SDG	Can	MPNet	Combined	OSDG	SDGi	KH	Aur	Full	AI/SDG	SDGi-VNR	UNGDC
SDG 1	15	7	15	12	17	1	8	15	12	17	15
SDG 2	11	11	11	11	10	7	6	11	10	13	11
SDG 3	10	4	10	8	7	8	9	10	11	7	9
SDG 4	13	14	13	13	16	13	12	13	13	14	14
SDG 5	16	17	16	16	14	15	15	16	16	16	17
SDG 6	4	6	4	3	5	4	1	4	2	5	5
SDG 7	3	1	3	4	3	2	7	3	3	3	3
SDG 8	14	16	14	14	11	9	16	14	14	15	12
SDG 9	17	15	17	15	13	17	14	17	17	12	16
SDG 10	8	10	8	2	6	10	11	8	9	6	10
SDG 11	7	3	7	7	9	11	5	7	8	9	2
SDG 12	9	8	9	9	12	12	2	9	6	10	4
SDG 13	2	9	2	1	2	5	3	2	5	2	7
SDG 14	5	5	5	5	1	3	4	5	7	4	8
SDG 15	6	12	6	6	4	6	10	6	4	8	6
SDG 16	12	13	12	10	15	14	13	12	15	11	13
SDG 17	1	2	1	–	8	16	17	1	1	1	1

5 Discussion

This section interprets the findings reported above. It moves from what the data show to what they mean: the structural pattern behind the two divergence dimensions, the robust patterns worth individual interpretation, and the implications and limits of the study. No new results are introduced here.

This dissertation makes two contributions, and they carry different weight. The first is a measurement framework: a way to separate coverage from framing, and to compare research and policy text directly in the same embedding space. This framework holds up. It survives changes of embedding model, reference source, and sampling, as the checks in Section 4.5 and the appendices show. The second is a set of findings about how AI research and SDG policy actually relate: which goals each side covers, how far apart their language sits within a goal, and whether the two move together across goals. These findings are weaker. The relationship between coverage gaps and semantic gaps reverses sign under some reference sources (Table 2), and SDG 17's position depends heavily on which centroid is used (Table 3). The framework should be read as the dissertation's main contribution. The findings about alignment should be read as a first application of it, worth checking again with a larger and more diverse policy corpus, not as a settled result.

5.1 Two Distinct Dimensions of Observed Divergence

The headline result of this study is simple: shared SDG labels are insufficient evidence of shared framing. Research attention alone is a poor predictor of semantic convergence. SDGs that attract substantial research attention can be either relatively close to policy framing (SDG 9) or substantially divergent from it (SDGs 3 and 7), while the strongest policy-dominant SDGs, especially SDGs 13 and 17, are also among the most semantically distant — a pattern that holds under the canonical and OSDG-derived centroids but reverses under Knowledge Hub and Aurora centroids (Table 2).

This pattern is best interpreted as a structural difference between knowledge systems rather than as a simple failure of communication. The Mode 1/2 distinction (Gibbons et al., 1994) sharpens this reading. The research corpus concentrates on technically tractable applications in health, energy, and infrastructure — SDGs 3, 7, and 9 — where dominant ML tasks align with well-defined problem-solving rather than institutional coordination, while the policy-side dominance of SDGs 13, 16, and 17 reflects Mode 2 goal-setting arising from contexts of application that demand coordination and collective action. Epistemic-community theory helps clarify the implication without overstating it: if expert influence partly depends on shared problem framing under uncertainty (Haas, 1992), then large within-SDG semantic gaps indicate where such shared framing may be less plausible. Following Cross (2013), however, semantic proximity is only a possible precondition for knowledge transfer; it cannot show whether expert communities exist, whether they have policy access, or whether policy actors use the relevant research. The two-communities theory (Caplan, 1979) reinforces this structural reading: the uncorrelatedness of coverage and semantic gap is what one would expect if the two corpora represent distinct knowledge cultures with different reward systems and knowledge-use criteria, rather than two sides of a shared enterprise.

This two-dimensional pattern is also robust to register adjustment (Appendix E): even aggressive controls that subtract the broad research-policy embedding direction compress but preserve the relative ranking of high-gap and low-gap SDGs. The most rigorous register-versus-substance separation attempt is the regression decomposition in Appendix E (Section E.5), which separates embedding variance associated with register from SDG structure.

The most robust aggregate finding is not about any single SDG: it is the dissociation between research attention and semantic divergence. As reported in Section 3.8, the H1c research-coverage test is near-zero and insensitive to excluding SDG 4, and the headline rankings survive corpus-subsampling (Appendix C) and register-adjustment (Appendix E) checks. The practical implication is that observed research–policy divergence is not a single scalar problem — attention allocation and framing divergence must be measured separately.

5.2 Robust Patterns of Divergence

The cross-sensitivity comparison narrows the set of gap findings that are stable enough to sustain individual interpretation. Only two SDGs maintain their extreme positions across all or nearly all measurement configurations: SDG 17 as the robustly highest-gap SDG, and SDG 9 as the robustly lowest-gap SDG. SDG 5 is also consistently among the lowest-gap goals across all checks.

SDG 17's top-gap position is stable under model substitution (rank 1 under MiniLM, rank 2 under MPNet), under all four policy sub-corpora (rank 1 in every family), and under OSDG-based centroids. Under alternative reference sources (SDGi-only, Knowledge Hub-only, Aurora-only), however, its rank varies considerably, reflecting the sensitivity of the combined centroid to source composition for this SDG. The claim that SDG 17 is the most divergent goal is therefore conditional on the combined reference source rather than instrument-independent.

This consistently largest gap is consistent with SDG 17's distinctive structural role. Le Blanc (2015) characterises SDG 17 as the dedicated goal for cross-cutting means of implementation, distinct from the substantive domains of the other 16 goals; the Griggs et al. (2017) guide similarly describes SDG 17 as providing the main enablers for implementing the entire SDG framework. Unlike every other SDG, where research and policy discourse both centre on a shared substantive topic, SDG 17 policy discourse concerns the implementation infrastructure for other goals — partnerships, finance, institutional coordination — a conceptual space that research abstracts on technical AI applications do not typically inhabit. The gap's stability across measurement conditions may therefore partly reflect this asymmetry between what the goal is (implementation enabler) and what the research corpus reports (technical substance), rather than divergent framing of shared content alone.

SDG 9 is the robustly closest SDG, occupying the bottom two or three ranks under every measurement configuration. Although this does not prove substantive alignment, it does demonstrate that the embedding-based instrument can separate a relatively close case from more visibly divergent ones.

The main coverage patterns are themselves robust. The full policy corpus is overwhelmingly dominated by the climate–partnerships–governance block (SDGs 13, 17, and 16). Appendix A.2 notes that this is source-family sensitive — the curated AI/SDG sub-corpus is more innovation-oriented — but the broader institutional profile holds across the UNGDC and SDGi families. The research corpus, conversely, concentrates on technically tractable SDGs (3, 7, 9, 15, 16).

A cluster of SDGs (6, 7, 13, 14) consistently occupy the upper half of the gap ranking across all checks, though their exact ordering shifts by no more than three positions. These SDGs are reliably above the median in research–policy divergence, but their precise rank is measurement-dependent and should not be over-interpreted individually. Similarly, the within-cluster spread observed between SDG 8 and SDG 10 within the 1/8/10 centroid cluster is a structural finding that does not depend on either SDG's precise rank: it undermines the concern that gap rankings are purely a byproduct of centroid similarity, because the instrument separates cluster members even where classifier leakage is strongest.

The remaining SDGs (1, 2, 3, 8, 10, 11, 12, 15, 16) shift position by four or more ranks under model substitution alone, with additional variation across reference and policy sources. No individual interpretive claims are made about their gap positions.

5.3 Implications

Two implications follow from the combined findings, and each pushes against the common habit of treating topical mention as evidence of alignment.

Research and policy agenda-setting. Rather than implying that funding should be mechanically redirected on the basis of embedding distances, the map indicates where research-policy translation, agenda-setting, and knowledge-brokerage work may be most needed. In Guston's terms, high-gap SDGs may be candidates for boundary-organising practices that create usable boundary objects while remaining accountable to both knowledge producers and policy users (Guston, 2001).

SDG-aware research communication. Simply increasing research output on neglected SDGs is insufficient; research communication norms may matter as well. Future work could test whether publication incentives that encourage explicit SDG framing improve semantic proximity more effectively than topic-level funding mandates alone. In practice, this could mean funder-required policy-facing summaries, journal-level SDG justification fields, or synthesis briefs written for policy audiences. Following Cash et al. (2003), this should be understood as boundary work rather than communication alone, requiring dual accountability to both scientific and policy principals (Guston, 2001).

These implications speak to distinct audiences. For SDG monitoring bodies, the finding that shared labels do not imply shared framing challenges keyword-based SDG mapping: coverage and framing should be reported as separate dimensions. For research funders, the results suggest that funding allocation alone is unlikely to close the semantic gap without complementary attention to how research communicates its policy relevance; portfolio-level semantic gap tracking could complement keyword-based counts by measuring whether funded research converges toward or diverges from policy discourse over successive cycles. For journals adopting SDG classification tags, the finding that framing divergence varies substantially by SDG recommends caution against treating SDG tagging as sufficient evidence of policy-aligned research.

5.4 Limitations

The cross-sensitivity comparison (Table 3) bounds which gap findings are stable enough to sustain interpretation. The limitations below address scope conditions that this comparison does not resolve.

Scope: semantic proximity, not substantive alignment or policy influence. The study measures semantic co-location in embedding space, not whether research and policy are substantively aligned or whether research is used by policy actors. Embedding proximity reflects shared vocabulary, not shared agenda, and does not measure policy influence, institutional mechanisms, or knowledge use — these would require methods beyond this study's scope.

Distance metric choice. This study uses cosine distance between mean embeddings. Other measures exist, such as Earth Mover's Distance or optimal transport between full point clouds rather than single centroids. These were not tested here. They may capture distributional shape that a single mean discards.

SDG 17 combined-source fragility. SDG 17 is the analysis's weakest measurement point. Its validation F1 is low across all centroid sources (0.300 combined, 0.437 SDGi-only, 0.316 Knowledge Hub, 0.267 Aurora), indicating the goal's broad mandate resists compact semantic definition. Additionally, the combined centroid's SDGi policy documents share genre with part of the scored policy corpus, which may marginally reduce the gap. The cross-sensitivity table (Table 3) confirms the concern: SDG 17 is rank 1 under the combined centroid but falls to rank 8 under SDGi-only centroids. Per-source validation is in Appendix A.1 (Table 4).

Corpus scope and retrieval boundaries. The policy corpus represents international institutional SDG discourse rather than the full universe of AI sustainability policy. Domestic plans, sub-national policies, and non-English documents are not represented; the research corpus is also English-only. Both biases likely understate Global South perspectives. Results are conditional on these boundaries. Appendix A.2 makes the source-family sensitivity explicit.

OpenAlex upstream classification. The research corpus is drawn from OpenAlex, whose SDG-topic tagging is itself a downstream classifier trained on external SDG-labelled data; the SDG assignments entering this study are therefore one step removed from human judgement and inherit any bias in that upstream model. Records that OpenAlex does not tag to any SDG are absent from the retrieval universe, so the corpus reflects OpenAlex's classification boundary rather than the full population of AI-for-sustainability research. This is a method-dependent corpus boundary in the sense of Armitage et al. (2020), and coverage proportions should be read as conditional on it.

Corpus asymmetry: AI-for-SDG scope. As noted in Section 3.2, the research corpus is AI $\cap$ SDG while the policy corpus is closer to a union of institutional SDG discourse and AI governance documents. Between-SDG rankings are conditional on this asymmetric construction.

Register effects on the semantic gap. Semantic distance may capture differences in communicative function (see Biber and Conrad (2009) on the register–genre–style distinction). Research abstracts and policy documents are text varieties designed to do different things, so the gap is a measure of observed textual-semantic divergence, not a purified estimate of substantive disagreement. Appendix E reports adjusted estimates; the spread between raw and aggressively adjusted values is best read as an uncertainty band around the register-sensitive component. The raw gap is retained because it best matches what the study measures.

Cross-sectional design. The study provides a snapshot of the 2018–2025 research period against a policy corpus assembled over a similar timeframe; it cannot address whether alignment is improving or deteriorating, or whether specific policy events (e.g., the EU AI Act 2024, IPCC AR6 2022) have shifted research framing.

6 Conclusion

This paper revisits an old problem with a new instrument. Policy scholars have long argued that research and policy often fail to meet one another on equal terms. What recent semantic tools add is not a final verdict, but the ability to map one important dimension of it directly, across thousands of texts.

The core finding is that shared SDG labels are insufficient evidence of shared framing. Research concentrates on SDGs 3, 4† (flagged as artefact-affected), and 9, while institutional policy discourse is dominated by SDGs 13, 16, and 17; within the same SDG, semantic divergence is largest in some of the most policy-dominant goals. The resulting map therefore separates cases where apparent alignment is strongest (SDG 9) from cases where shared labels conceal substantial divergence (SDGs 13 and 17, though SDG 17's gap estimate is conditional on the combined reference source; see Section 5.4). This contrast is the dissertation's main conceptual contribution: embedding-based distance can help identify where closer investigation is warranted, not as a final verdict on any given gap. The framework, not any single gap estimate, is the main product of this work.

These findings are bounded by several limitations detailed in Section 5.4: the semantic gap reflects register differences and corpus asymmetries, and the SDG 17 estimate is particularly fragile to reference-source composition. A natural extension would track whether semantic gaps narrow or widen across successive policy cycles as further AI governance frameworks emerge. A further extension would repurpose the centroid distances computed here to map which SDGs sit close to or depend on one another in semantic space: because the goals are interdependent by design (Nilsson et al., 2016), the same embedding space that compares research and policy could also chart SDG interlinkages more finely than current network-based accounts.

The central implication is not that the AI-for-sustainability field is aligned or misaligned in aggregate — it is both, depending on which SDG and which dimension one examines. Coverage and framing are separate axes of divergence; a field that appears well-aligned on one axis may be substantially divergent on the other. Semantic mapping cannot replace qualitative understanding of why particular gaps persist or what they mean for policy uptake, but it can show — at scale and with transparent assumptions — where the questions are worth asking.

More broadly, this dissertation argues that alignment should be understood as a multi-level empirical construct — from lexical overlap through semantic proximity to substantive agreement (Section 3.1) — rather than as a binary property inferred from shared labels.

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A Diagnostics for Centroid Construction

A.1 Per-SDG Reference-Source Comparison

The canonical centroids combine OSDG, SDGi, Knowledge Hub, and Aurora texts per SDG. This appendix compares each single-source centroid to the canonical combined centroid across four metrics: validation F1, centroid cosine similarity, research coverage (papers assigned per SDG), and within-SDG semantic gap. The purpose is to assess whether any single source dominates the combined centroid and whether the study's headline results are sensitive to the choice of reference corpus.

Table 4 reports per-SDG validation F1 and centroid similarity for all four configurations:

- **Combined** (canonical): all available single-label texts (OSDG + SDGi + Knowledge Hub) per SDG.
- **OSDG-only**: the OSDG Community Dataset only (SDGs 1–16; SDG 17 is not available in OSDG).
- **SDGi-only**: the SDGi Corpus only (VNR/VLR policy reports).
- **Knowledge Hub-only**: the IISD SDG Knowledge Hub only (journalism).

Table 4. Per-SDG source comparison: validation F1 and centroid cosine similarity to the canonical combined centroid.

SDG	Combined		OSDG		SDGi		Knowledge Hub		Aurora	
	F1	cos	F1	cos	F1	cos	F1	cos	F1	cos
SDG 1	0.687	1.000	0.697	0.990	0.708	0.914	0.000	0.512	0.522	0.823
SDG 2	0.854	1.000	0.844	0.985	0.780	0.898	0.792	0.810	0.567	0.800
SDG 3	0.680	1.000	0.583	0.959	0.607	0.872	0.678	0.732	0.630	0.879
SDG 4	0.907	1.000	0.920	0.993	0.879	0.897	0.857	0.703	0.872	0.798
SDG 5	0.737	1.000	0.712	0.996	0.658	0.890	0.649	0.725	0.638	0.843
SDG 6	0.878	1.000	0.869	0.993	0.839	0.898	0.848	0.845	0.650	0.785
SDG 7	0.878	1.000	0.871	0.995	0.869	0.906	0.860	0.847	0.882	0.904
SDG 8	0.548	1.000	0.523	0.983	0.407	0.893	0.362	0.748	0.327	0.818
SDG 9	0.678	1.000	0.618	0.992	0.615	0.850	0.552	0.730	0.160	0.757
SDG 10	0.733	1.000	0.712	0.985	0.600	0.874	0.150	0.525	0.108	0.618
SDG 11	0.545	1.000	0.509	0.991	0.571	0.898	0.630	0.749	0.538	0.892
SDG 12	0.684	1.000	0.701	0.981	0.699	0.946	0.659	0.877	0.508	0.869
SDG 13	0.694	1.000	0.644	0.986	0.623	0.915	0.604	0.900	0.227	0.774
SDG 14	0.822	1.000	0.822	0.982	0.789	0.917	0.784	0.836	0.613	0.800
SDG 15	0.683	1.000	0.683	0.992	0.699	0.933	0.667	0.861	0.571	0.875
SDG 16	0.870	1.000	0.845	0.995	0.800	0.796	0.780	0.691	0.781	0.927
SDG 17	0.300	1.000	–	–	0.437	0.937	0.316	0.982	0.267	0.821

Notes: A "–" indicates no texts for that SDG in the source (OSDG, SDG 17) or a trivial self-similarity of 1.0.

The bootstrap applies to the validation benchmark only. That benchmark is a small, fixed sample, and the bootstrap tests whether its F1 differences reflect real separation or

Table 5. Per-SDG source comparison: research coverage and within-SDG semantic gap for each reference source.

SDG	Combined		OSDG		SDGi		Knowledge Hub		Aurora	
	n	gap	n	gap	n	gap	n	gap	n	gap
SDG 1	11,804	0.253	18,348	0.329	14,782	0.266	35,146	0.805	21,489	0.447
SDG 2	83,441	0.324	88,364	0.334	100,220	0.344	160,806	0.449	120,798	0.505
SDG 3	470,627	0.423	461,638	0.442	421,276	0.413	425,281	0.436	377,884	0.434
SDG 4	355,720	0.287	353,269	0.287	367,183	0.289	382,814	0.298	415,692	0.317
SDG 5	53,741	0.249	58,680	0.254	81,372	0.297	71,840	0.275	49,479	0.257
SDG 6	90,715	0.491	104,730	0.514	72,936	0.430	129,462	0.522	119,951	0.595
SDG 7	198,743	0.497	216,430	0.503	233,924	0.512	340,638	0.568	150,021	0.497
SDG 8	41,898	0.269	38,662	0.286	63,828	0.317	124,609	0.379	88,471	0.241
SDG 9	247,370	0.234	271,741	0.261	240,883	0.314	131,436	0.236	297,119	0.288
SDG 10	20,323	0.432	25,249	0.516	42,974	0.429	76,100	0.372	85,597	0.328
SDG 11	136,616	0.447	141,142	0.466	91,158	0.346	108,151	0.368	190,345	0.525
SDG 12	90,960	0.424	96,433	0.430	62,947	0.316	97,454	0.355	66,345	0.577
SDG 13	110,645	0.528	155,820	0.572	141,882	0.529	62,480	0.495	160,148	0.551
SDG 14	99,162	0.485	101,402	0.492	123,624	0.534	115,658	0.527	100,242	0.545
SDG 15	175,447	0.465	194,906	0.472	214,340	0.504	186,868	0.485	107,125	0.427
SDG 16	189,046	0.323	216,884	0.368	174,806	0.297	69,880	0.289	133,242	0.317
SDG 17	167,440	0.671	0	–	95,563	0.406	25,075	0.255	59,750	0.191

Notes: Coverage is the count of research papers assigned to each SDG via nearest-centroid matching. Gap is $1 - \cos(\text{research centroid}, \text{policy centroid})$; larger values indicate greater research–policy divergence. Each row compares the same SDG's gap across the four reference sources; read down a column to see how a source shifts the gap, and across a row to see whether the gap ranking is stable.

sampling noise. Coverage and semantic gap estimates are computed over the full research and policy corpora, not a sample from a larger population, and their stability is tested separately in Appendix C and the cross-sensitivity checks in Table 3 and Appendix A.2.

Several patterns emerge from Table 4. First, OSDG-only centroids are very close to the combined direction for all 16 available SDGs (cosine 0.959–0.996), and their validation F1 scores match the combined within 0.05 in most cases. This is expected: OSDG contributes the largest text count per SDG (718–3,401 texts), so the combined centroid is OSDG-dominated for SDGs 1–16.

Second, the supplementary sources (SDGi, Knowledge Hub) introduce modest centroid shifts, with cosine to combined ranging from 0.797 (SDG 16, SDGi) to 0.945 (SDG 12, SDGi) for SDGi, and from 0.512 (SDG 1, KH) to 0.900 (SDG 13, KH) for Knowledge Hub. The largest deviations occur for SDG 1 (KH cosine 0.512) and SDG 10 (KH cosine 0.525), suggesting that these SDGs have heterogeneous semantic scope across policy and journalism genres. These sources have fewer texts (18–364 per SDG) and pull the centroid in different genre-specific directions.

Third, the supplementary sources do not degrade the combined F1 for SDGs 1–16. For most of these SDGs, the combined F1 is close to the best single-source F1. In several

Table 6. Bootstrap 95% confidence intervals for the canonical (combined MiniLM) validation F1, per SDG and macro.

SDG	F1	95% CI low	95% CI high
SDG 1 (No Poverty)	0.680	0.535	0.800
SDG 2 (Zero Hunger)	0.852	0.766	0.927
SDG 3 (Good Health)	0.671	0.500	0.808
SDG 4 (Quality Education)	0.906	0.835	0.966
SDG 5 (Gender Equality)	0.735	0.613	0.840
SDG 6 (Clean Water)	0.875	0.800	0.936
SDG 7 (Clean Energy)	0.876	0.800	0.938
SDG 8 (Decent Work)	0.542	0.381	0.690
SDG 9 (Industry & Infra.)	0.676	0.528	0.800
SDG 10 (Reduced Inequalities)	0.730	0.590	0.851
SDG 11 (Sustainable Cities)	0.544	0.378	0.706
SDG 12 (Responsible Cons.)	0.681	0.545	0.800
SDG 13 (Climate Action)	0.690	0.567	0.810
SDG 14 (Life Below Water)	0.819	0.710	0.900
SDG 15 (Life on Land)	0.677	0.552	0.786
SDG 16 (Peace & Justice)	0.868	0.767	0.944
SDG 17 (Partnerships)	0.297	0.148	0.448
Macro-F1 (mean)	0.713	0.680	0.748

Notes: 1,000 resamples with replacement over the 616-text benchmark; predictions are fixed by the frozen centroids, so each resample redraws (true, predicted) instance pairs. SDGs absent from a resample score 0. The middle band (F1 $\approx$ 0.67–0.69) is not separated from the macro mean or from itself.

cases (SDGs 8, 11, 13), the combined F1 exceeds every single-source F1, confirming that combining sources broadens semantic coverage in a way that improves discriminability rather than diluting it. SDG 17 is a notable exception: the combined F1 (0.300) is lower than either single-source alternative. This is because the combined centroid (1,044 texts, 59% Knowledge Hub, 31% SDGi, 10% Aurora) inherits the KH direction, which does not align optimally with the benchmark's policy-excerpt genre, while the SDGi-only centroid (321 texts) benefits from within-genre overlap with the benchmark.

Table 5 shows that the coverage and gap findings are also qualitatively robust to source choice. The SDG ranking by coverage is largely preserved across all four sources: SDGs 3, 4, 7, 9, 15, and 16 consistently appear among the largest clusters, while SDGs 1, 10, and 12 are consistently among the smallest. The Spearman rank correlation between combined and OSDG-only coverage is near-perfect ($\rho > 0.98$), and between combined and SDGi or KH it remains above 0.85. The semantic gap patterns are similarly stable: SDG 17 has the largest gap under every source except OSDG (where it is unavailable), and the top-three gap SDGs (17, 13, 10–6) are consistent across Combined, OSDG, and SDGi. The KH-only centroids produce larger gap variability, particularly for SDG 1 (0.805) and SDG 10 (0.372), reflecting the small text counts in those SDGs for the journalism corpus.

For SDGs 1–16, the combined centroid is a balanced instrument: it is not dominated by any single source and achieves comparable or better validation F1 than any single-source alternative, while preserving the coverage and gap patterns reported in the main analysis. For SDG 17, where OSDG coverage is absent, the combined centroid is a conservative choice — it does not overfit to any single annotation genre, but it reflects a compromise direction that is less discriminative on the benchmark than either single-source alternative.

A.2 Policy Source-Family Sensitivity

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Tables 7 and 8 therefore asks a narrow question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The broad international-institutional profile is clearly source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the SDGi and UNGDC families are more strongly concentrated on climate, partnerships, governance, and, in the SDGi case, SDG 11-style institutional review language. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and SDG 17 remains the largest semantic-gap case across all three families.

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. The scope of what the policy-side distribution represents is accordingly narrower, but the main diagnostic purpose remains valid.

A.3 SDGi Circularity Check

The SDGi Corpus supplies part of the SDG centroids (Section 3.5) and also supplies 31,971 of the 54,082 policy segments (59.1%) scored against those centroids. This overlap could inflate the policy-side score and explain part of the 0.328 policy–research gap (Section 3.9).

To test this, I rebuilt the centroids from OSDG, Knowledge Hub, and Aurora only, with SDGi removed. I then rescored the SDGi policy segments against this SDGi-excluded centroid. The mean top score was 0.583, against a canonical policy mean of 0.560. The research-paper mean top score against the same centroid was 0.246, against a canonical 0.232. The resulting gap was 0.337, against the canonical 0.328 — a difference of 0.009.

Rescoring the full policy corpus against the SDGi-excluded centroid gives a mean top

Table 7. Per-SDG policy source-family sensitivity: coverage share.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	n	share (%)	n	share (%)	n	share (%)	n	share (%)
SDG 1	2,970	1.8	1,158	4.2	1,719	8.9	93	0.6
SDG 2	2,119	0.8	360	1.1	1,665	3.8	94	0.3
SDG 3	3,893	0.4	1,127	4.2	2,699	1.6	67	0.0
SDG 4	3,009	0.2	391	0.0	2,556	1.3	62	0.0
SDG 5	3,107	0.8	364	0.0	2,532	0.3	211	0.9
SDG 6	1,992	0.4	231	0.0	1,727	1.9	34	0.2
SDG 7	2,563	0.4	617	0.0	1,859	1.0	87	0.3
SDG 8	2,623	1.0	653	0.0	1,953	7.0	17	0.1
SDG 9	4,601	2.4	2,907	45.3	1,625	1.9	69	0.5
SDG 10	852	0.0	387	0.0	447	0.0	18	0.0
SDG 11	2,797	2.7	283	0.0	2,507	21.4	7	0.0
SDG 12	2,283	0.4	605	0.0	1,665	3.5	13	0.0
SDG 13	4,972	25.0	1,227	18.9	1,781	3.2	1,964	28.7
SDG 14	2,058	0.2	621	0.0	1,275	0.3	162	0.1
SDG 15	2,395	0.7	573	0.0	1,730	0.6	92	0.7
SDG 16	3,769	14.5	1,101	2.1	1,481	0.3	1,187	17.3
SDG 17	8,079	48.2	3,034	24.2	2,750	42.8	2,295	50.1

Notes: For each policy source family, the table reports the number of policy segments assigned to each SDG and the document-weighted share (%) of that family's documents assigned to each SDG.

score of 0.556, close to the canonical 0.560.

The gap does not shrink when SDGi is removed from the centroid. Since SDGi is 59.1% of the policy corpus, a real circularity effect would likely have shown up here. It did not. This is a null result: the policy–research score asymmetry is not explained by the overlap between the SDGi centroid source and the SDGi policy segments.

A.4 SDG 4 Lexical Artefact Audit

The SDG 4 concern is not used to relabel research records. Its narrower purpose is to test whether the concern is empirically plausible. Table 9 therefore compares the lexical composition of SDG 4-assigned research records with a matched non-SDG4 sample and with SDG 9, using only two transparent keyword groups: ML-learning vocabulary and education-specific vocabulary.

The audit points to a more nuanced conclusion than a pure false-positive story. More than half of the SDG 4-assigned set contains education-specific vocabulary without ML-only terms, and roughly another third contains both education and ML-learning vocabulary. By contrast, the matched non-SDG4 sample and the SDG 9 comparison set are dominated by ML-only vocabulary. This means the SDG 4 signal is not simply the same technical language that drives SDG 9. However, the large "both" share also confirms that ML-

Table 8. Per-SDG policy source-family sensitivity: semantic gap.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	n	gap	n	gap	n	gap	n	gap
SDG 1	2,214	0.253	402	0.312	1,719	0.253	93	0.403
SDG 2	2,020	0.324	261	0.393	1,665	0.319	94	0.477
SDG 3	3,506	0.424	740	0.320	2,699	0.463	67	0.568
SDG 4	2,994	0.287	376	0.287	2,556	0.298	62	0.448
SDG 5	3,066	0.249	323	0.254	2,532	0.257	211	0.343
SDG 6	1,941	0.491	180	0.531	1,727	0.489	34	0.619
SDG 7	2,343	0.497	397	0.521	1,859	0.494	87	0.637
SDG 8	2,415	0.268	445	0.286	1,953	0.276	17	0.472
SDG 9	3,626	0.234	1,932	0.205	1,625	0.368	69	0.395
SDG 10	732	0.433	267	0.428	447	0.478	18	0.567
SDG 11	2,735	0.446	264	0.441	2,464	0.455	7	0.680
SDG 12	1,951	0.425	273	0.473	1,665	0.429	13	0.625
SDG 13	4,530	0.529	785	0.478	1,781	0.510	1,964	0.594
SDG 14	1,675	0.486	238	0.467	1,275	0.493	162	0.586
SDG 15	2,156	0.465	334	0.507	1,730	0.461	92	0.599
SDG 16	3,408	0.324	740	0.273	1,481	0.378	1,187	0.460
SDG 17	6,651	0.671	1,609	0.591	2,747	0.697	2,295	0.739

Notes: For each policy source family, the table reports the number of capped policy segments and the within-SDG research–policy semantic gap. Each row compares the same SDG's gap across the three source families and the full corpus; read across a row to see whether the gap is stable or driven by one family.

learning language and education language overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical nearest-centroid assignment but continues to treat SDG 4 as artefact-affected. The most accurate reading is a learning-related assignment whose interpretation is blurred by lexical overlap — neither "clean education" nor "pure ML artefact."

Table 9. SDG 4 lexical artefact audit.

Corpus subset	N	ML-only %	Education-only %	Both %	Neither / unclear %
SDG 4-assigned research	355720	7.3	51.9	31.2	9.5
Non-SDG4 research sample	355720	55.9	3.9	3.1	37.1
SDG 9-assigned research	247370	52.0	4.1	4.0	39.9

Notes: The audit is not used to reassign SDGs. It only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

A.5 Centroid Similarity

This subsection collects the two diagnostic views of how the 17 SDG centroids relate to one another. The first is the pairwise cosine-similarity heatmap (Figure 4); the second is the full benchmark confusion matrix (Table 10) that underlies the per-SDG F1 scores reported in Table 1. Together they let the reader check, for any pair of goals, both how close the centroids sit in embedding space and how often the instrument actually swaps one label for another on the 616 benchmark texts.

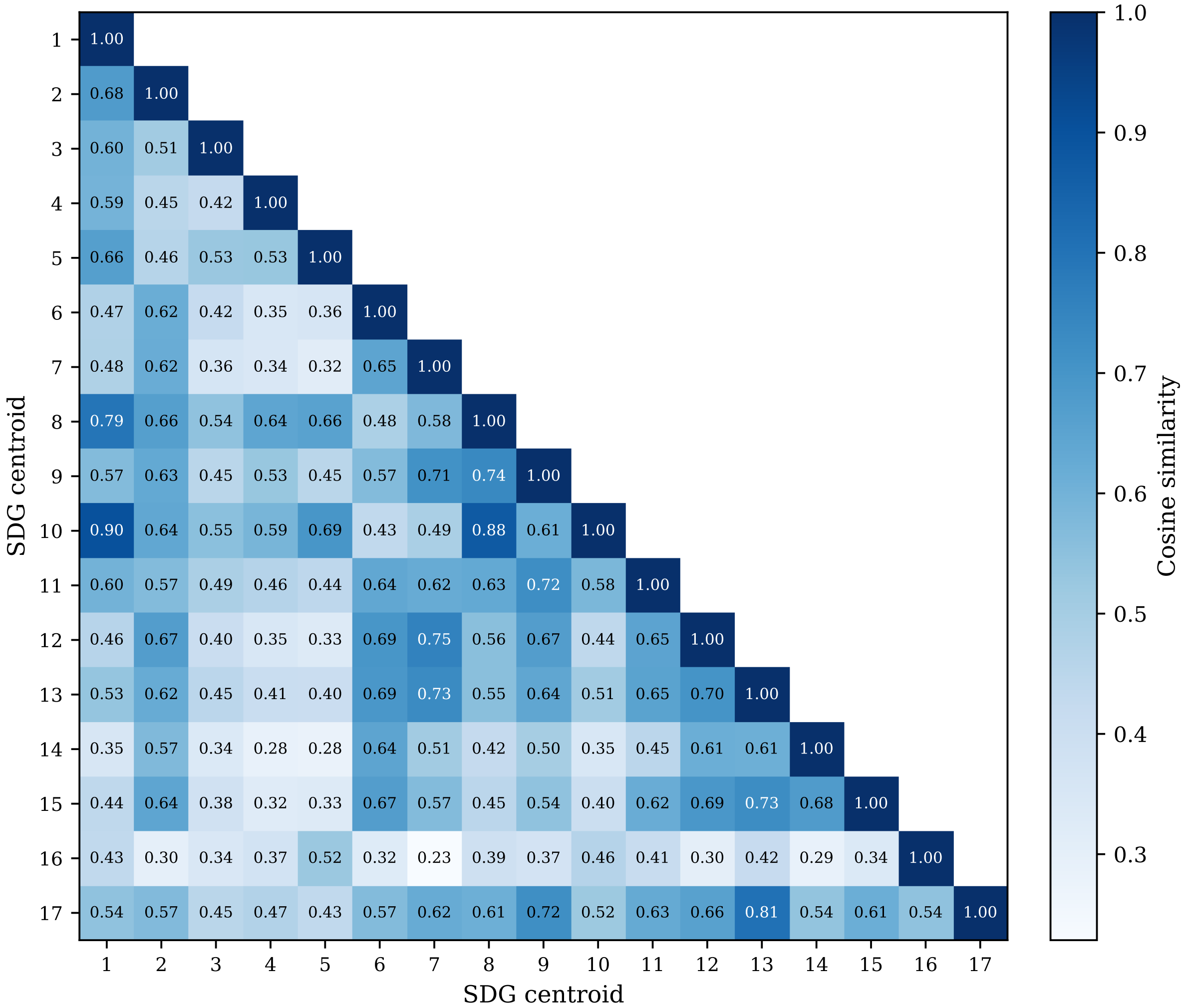


Figure 4. Pairwise cosine similarity between canonical SDG centroids (lower triangle). SDG 13 and SDG 17 sit close together (cosine 0.805), as do SDGs 1, 8, and 10.

Figure 4 shows the full 17×17 similarity grid. The lower triangle is annotated; the upper triangle is left blank because the matrix is symmetric. Reading the grid, three observations stand out. First, the diagonal dominates: most centroids are far more similar to themselves than to any neighbour, which is why the instrument reaches a macro-F1 of 0.716 (Table 1) with no supervised training. Second, a small set of off-diagonal cells is elevated. SDG 11's centroid neighbours SDG 9 (cosine 0.720), and SDG 8 lies near SDGs 1 and 10 (cosine 0.903 and 0.794) — these are the proximities that make the SDG 11 and SDG 8 scores in Section 4.1 indicative rather than exact. Third, SDG 17 is the clear outlier: its row and column are the flattest, and its nearest neighbour is SDG 13 (cosine 0.805), reflecting the overlap between climate and partnership discourse noted in Section 4.1.

Table 10. Benchmark confusion matrix (n = 616). Rows are true SDGs, columns are predicted SDGs. The diagonal is the count of correct assignments; off-diagonal cells are misassignments. This matrix underlies the per-SDG F1 scores in Table 1.

True ↓ / Pred →	SDG1	SDG2	SDG3	SDG4	SDG5	SDG6	SDG7	SDG8	SDG9	SDG10	SDG11	SDG12	SDG13	SDG14	SDG15	SDG16	SDG17
SDG SDG1	23	0	0	0	0	0	0	1	0	2	0	0	0	0	0	1	0
SDG SDG2	1	38	0	0	0	1	0	0	1	0	0	0	0	0	4	0	0
SDG SDG3	2	0	17	0	2	0	0	0	0	0	3	2	0	0	0	0	2
SDG SDG4	1	0	0	39	0	0	0	0	0	2	0	0	0	0	0	0	1
SDG SDG5	1	0	3	0	28	0	0	2	0	0	0	0	0	0	0	1	0
SDG SDG6	0	0	0	0	0	43	1	0	0	0	1	0	0	2	1	0	0
SDG SDG7	2	1	0	1	0	0	43	0	0	0	1	1	0	0	1	0	0
SDG SDG8	2	1	1	0	5	1	0	17	2	2	3	0	1	0	0	1	2
SDG SDG9	0	0	0	1	2	1	0	0	20	1	1	0	0	0	0	0	2
SDG SDG10	3	1	0	0	1	0	0	1	0	22	0	0	0	0	0	1	0
SDG SDG11	1	0	0	0	0	2	0	0	0	1	15	2	4	0	1	0	1
SDG SDG12	1	1	1	1	0	1	2	0	0	0	0	26	2	0	2	0	6
SDG SDG13	0	0	0	0	0	0	0	0	1	0	2	0	25	1	1	0	3
SDG SDG14	0	0	0	0	0	0	0	0	1	0	0	1	0	30	2	0	1
SDG SDG15	1	2	0	0	1	1	0	0	0	0	1	0	0	5	28	0	2
SDG SDG16	0	0	0	1	1	0	0	2	0	0	0	0	1	0	0	30	0
SDG SDG17	2	0	0	0	1	0	2	1	6	1	1	1	6	0	1	0	9

Notes: Each of the 616 benchmark texts is assigned to the nearest centroid; rows sum to the per-SDG true counts, columns to predicted counts.

Table 10 reports every assignment, so the heatmap's proximity claims can be checked against actual errors. The diagonal confirms the macro-F1 result: most goals are assigned correctly in the large majority of cases. The off-diagonal mass is concentrated and sparse. The largest single confusion is SDG 15 → SDG 14 (5 of 30 misassigned SDG 15 texts), with SDG 2, SDG 17, and others taking one or two each; in the reverse direction, only 2 of 30 misassigned SDG 14 texts land on SDG 15. SDG 11 and SDG 8, whose centroids sit near neighbours in Figure 4, do not in fact bleed heavily into those neighbours in the confusion matrix — their weak F1 comes more from diffuse spread across several goals than from one dominant swap. SDG 17 again shows the weakest separation: across its 30 benchmark texts the predicted labels scatter most widely, consistent with its status as the lowest-F1 goal in Table 1. The matrix therefore corroborates the two claims made in Section 4.1: the instrument is reliable for most goals, and the uncertainty is concentrated in a few cross-cutting or semantically adjacent pairs rather than spread evenly across all 17.

B Supplementary Robustness and Sensitivity Checks

B.1 Combined Research–Policy PCA Landscape

Before presenting the main alignment metrics, I also visualise the empirical semantic landscape of the corpus using PCA. Research-paper embeddings and policy-text embeddings are first combined into a shared corpus and projected into two principal components. This projection functions as a descriptive diagnostic of how the two domains occupy the broader embedding space, not as an inferential measurement. I then overlay the 17 SDG reference centroids, derived from labelled SDG benchmark data, using the same fitted PCA transformation. This places the SDG anchors in relation to the actual distribution of research and policy discourse. Where research and policy clouds overlap around an SDG anchor, the visualisation suggests shared semantic framing; where the research and policy SDG centroids occupy distinct regions, it provides an intuitive illustration of the semantic gaps later measured through cosine distances in the original 384-dimensional space.

Because the research corpus is much larger than the policy corpus, PCA is fitted on a balanced sample consisting of all usable policy segments and an equal-sized random sample of research papers. In the current canonical run, this means 54,082 policy segments and 54,082 research papers drawn from totals of 54,082 policy segments and 2,543,698 research papers. This prevents the visual projection axes from being dominated by variance in the research corpus. The first two principal components explain only 7.2% and 3.5% of variance respectively, so the figure is a coarse descriptive map, not a precise geometric summary of 384-dimensional distances. The combined PCA also appears strongly shaped by broad research–policy separation, which is exactly why it is retained as a diagnostic rather than as a measurement device. The full corpus is retained for the main quantitative analysis, and all reported cosine-distance metrics are calculated in the original 384-dimensional Sentence-BERT embedding space.

The visual placement of the SDG anchors also warrants careful interpretation. Because the reference centroids are derived from benchmark and institutional SDG text, they may naturally sit closer to policy-style language than to research title-abstract texts. The caution is against reading two-dimensional proximity to the green anchors as direct proof of substantive alignment, not against the centroid method itself.

B.2 Within-Corpus SDG Structure Diagnostics

To assess whether the SDG-centroid method produces meaningful structure within each corpus, I also visualise research and policy embeddings separately using corpus-specific PCA projections. In each case, PCA is fitted only within that corpus, documents are assigned to their nearest SDG centroid using the existing scoring procedure, and corpus-specific SDG centroids are overlaid. These plots are diagnostic only: they show whether SDG assignments form relatively coherent regions or whether the corpus appears as heavily overlapping semantic fog. Because PCA is fitted separately for research and policy, the two panels do not share a common coordinate system and should not be compared spatially.

I pair the visual diagnostics with centroid-coherence and clustering statistics saved alongside the figure outputs. In the current canonical run, the research panel plots 100,000 sampled documents and the policy panel plots 54,082 segments. Sample-based cosine silhouette scores are 0.023 for research and 0.072 for policy. The figures show that research assignments

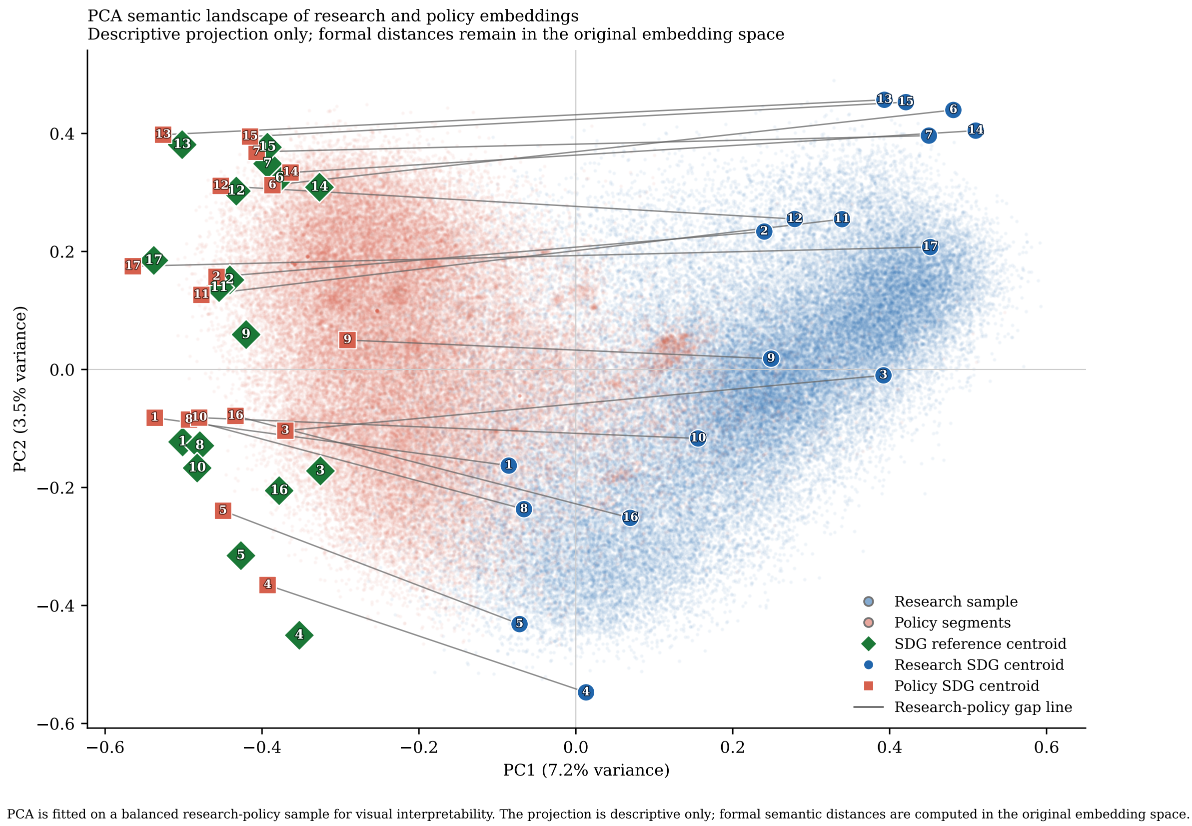


Figure 5. PCA semantic landscape of the research-policy embedding space. Background points show a balanced research-policy sample used to fit and visualise the two-dimensional projection. Diamond markers show the 17 SDG reference centroids. Circular and square markers show the full-corpus research and policy SDG centroids respectively, with thin line segments linking each research-policy pair.

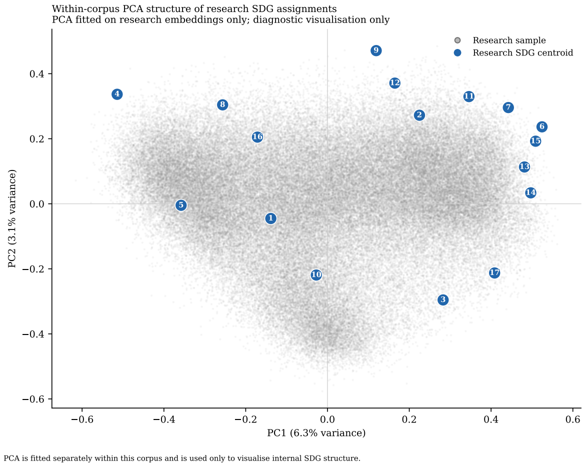
Notes: The projection is descriptive only; formal semantic distances are computed in the original embedding space.

are especially fuzzy and overlapping, while policy exhibits somewhat more internal structure without approaching cleanly separated clusters. These visuals therefore neither validate nor invalidate the 384D centroid method by themselves. They simply show that real-world SDG discourse is semantically fuzzy, especially in the research corpus, which is why the main analysis treats the centroid procedure as a transparent anchoring device rather than as a definitive classifier.

B.3 Lexical Illustration of the Semantic Gap

Table 11 adds a small lexical interpretability check for two high-gap, policy-dominant cases (SDGs 17 and 13) and one lower-gap comparison case (SDG 9). The table is generated from deterministic samples of existing hard-assigned research and policy texts. It reports distinctive terms rather than selected quotations, so it is a descriptive guide to language patterns, not qualitative coding or proof of policy intent.

The contrast helps make the embedding distances more legible. In SDGs 17 and 13, the



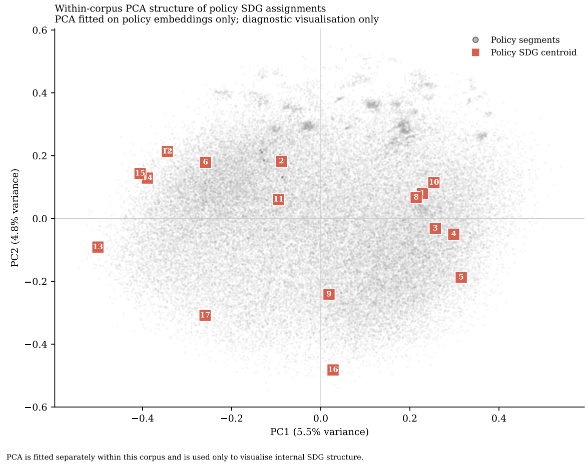


Figure 6. Within-corpus PCA diagnostics for SDG assignments. Left: research embeddings with corpus-specific research SDG centroids overlaid. Right: policy embeddings with corpus-specific policy SDG centroids overlaid.

Notes: PCA is fitted separately within each corpus and is used only to visualise internal SDG structure. The two panels do not share a coordinate system. Quantitative diagnostics are reported alongside the visualisation.

high semantic gaps correspond to a split between technical research language and more institutional, international, national-government, and climate-action policy language. SDG 17 is especially cautionary: its research-side terms are biomedical/ML-heavy, while its policy-side terms are international and agenda-oriented. SDG 9, by contrast, shows more overlap around technology, infrastructure, innovation, and systems language. This does not prove substantive alignment in SDG 9 or substantive failure in SDGs 13 and 17; it simply illustrates why shared SDG labels can conceal different textual framings.

B.4 Softmax Multi-label SDG Membership

The canonical analysis forces each text into exactly one SDG by nearest-centroid assignment. This is transparent and directly interpretable, but it also imposes a one-SDG-per-text assumption on discourse that is naturally multi-label. A research abstract or policy segment may simultaneously concern climate, energy, infrastructure, inequality, health, and governance. To stress-test this assumption, I implement an alternative specification in which each text contributes fractionally to all 17 SDGs.

For each text, cosine similarities to the 17 SDG reference centroids are transformed into temperature-scaled softmax weights (softmax is a function that converts any set of scores into a probability-like distribution that sums to 1), so the weights sum to one across SDGs. I test both a raw softmax and a corpus-calibrated variant that standardises similarities within each corpus before applying the softmax. The calibrated version is designed to reduce reliance on absolute proximity to benchmark-derived centroids, which may partly reflect policy-style language, but it does not remove benchmark or register bias. These

Table 11. Lexical interpretability check for selected semantic-gap cases.

SDG	Gap	Research-side distinctive terms	Policy-side distinctive terms	Descriptive reading
SDG 17	0.671	model, learning, language, neural, deep, cells, analysis, method	countries, nations, international, united, global, agenda, country, world	High-gap case: the research-side sample is technical and biomedical/ML-heavy, while policy language is international, institutional, and agenda-oriented.
SDG 13	0.528	model, learning, neural, prediction, method, machine, network, deep	climate, change, emissions, countries, national, global, action, nations	High-gap case: both sides concern climate, but policy language is more institutional and commitment-oriented while research language is more technical and measurement-oriented.
SDG 9	0.234	learning, machine, analysis, model, network, neural, detection, iot	national, government, infrastructure, public, countries, country, sector, strategy	Lower-gap comparison: differences remain visible, but the contrast is closer to a technology/infrastructure register split than to the broader institutional gap seen in SDGs 13 and 17.

Notes: Terms are generated from deterministic samples of research and policy texts assigned to each SDG. The table is descriptive only and is not used to relabel records or replace the embedding-based semantic-gap estimates.

variants are therefore treated as stress tests rather than repairs.

The results support a cautious but informative interpretation. Low-temperature raw softmax broadly preserves the hard-label structure: at $T = 0.03$, the rank correlation between hard and soft coverage gaps is 0.944, the corresponding semantic-gap Spearman correlation is 0.819, and the mean absolute differences are 0.024 for coverage and 0.053 for semantic gap. At the much softer setting $T = 0.20$, those quantities move to 0.782, 0.716, 0.068, and 0.167 respectively, showing that diffuse membership eventually changes both coverage and semantic-gap rankings more substantially.

The corpus-calibrated variant is the stronger stress test. Across the tested temperatures, its coverage-gap Spearman correlation remains in the range 0.792–0.809, while the semantic-gap Spearman correlation ranges from 0.767 to 0.816. Mean absolute differences stay around 0.054–0.057 for coverage and 0.056–0.064 for semantic gap. However, the top-three semantic-gap leaders have zero overlap with the hard-label baseline at every tested temperature, which means the calibrated variant materially changes the exact SDG ranking of the most divergent cases.

Table 12. Softmax-weighted multi-label robustness summary.

Variant	T	ρ cov.	ρ sem.	MAD cov.	MAD sem.	Top-3 sem. overlap
Raw softmax	0.03	0.944	0.819	0.024	0.053	2/3
Raw softmax	0.05	0.926	0.708	0.031	0.083	2/3
Raw softmax	0.10	0.885	0.681	0.049	0.132	1/3
Raw softmax	0.20	0.782	0.716	0.068	0.167	1/3
Calibrated softmax	0.03	0.809	0.767	0.054	0.056	1/3
Calibrated softmax	0.05	0.809	0.767	0.054	0.057	1/3
Calibrated softmax	0.10	0.806	0.806	0.055	0.058	1/3
Calibrated softmax	0.20	0.792	0.816	0.057	0.064	1/3

Notes: Spearman correlations compare hard-label and softmax-based SDG rankings. Mean absolute differences (MAD) are absolute deviations from the hard-label baseline. The final column shows overlap in the top three semantic-gap SDGs relative to the hard-label ranking.

The evidence points in two directions: low-temperature raw softmax broadly preserves the hard-label structure, suggesting the main findings are not artefacts of winner-takes-all assignment; but the corpus-calibrated variant materially changes exact rankings, meaning broad patterns should be emphasised over exact SDG rank ordering.

B.5 Implications for Interpreting the Main Estimates

These diagnostics are methodological stress tests that clarify the limits of the centroid approach, rather than failed repairs. The PCA diagnostics show that real-world SDG discourse is fuzzy rather than cleanly clustered, especially in the research corpus. The SDG 17 sparse-reference check shows that the headline SDG 17 result is not solely a one-resample accident, while still preserving the qualification that SDG 17 rests on a much smaller benchmark base than the other goals. The softmax multi-label analysis shows that the hard-label baseline is broadly stable when the one-SDG-per-text assumption is softened, but that corpus-calibrated variants can change exact rankings. The policy source-family split shows that the full-corpus policy profile is a broader institutional SDG discourse rather than a universal AI policy profile. The SDG 4 audit shows that the concern is real, but more nuanced than a pure machine-learning false positive. The canonical hard-label method is therefore retained because it is transparent, reproducible, and directly interpretable. The alternative specifications are reported to document boundary conditions rather than to replace the primary quantity of interest.

These analyses also show that the centroid method should be interpreted as a transparent semantic anchoring procedure rather than a definitive operational SDG classifier. Attempts to address multi-label overlap and corpus-register bias improve some aspects of the design but introduce new modelling choices. For this dissertation, the hard-label centroid method is therefore retained as the canonical specification, while the alternatives are reported as robustness and boundary-condition checks.

C Sample-Stability Robustness Check

To test whether the headline findings depend heavily on research-corpus size, I reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), I drew 100 independent samples without replacement, recomputed the research coverage profile, rebuilt sampled research centroids, recalculated the mean within-SDG semantic gap, and re-estimated the policy-text calibration-bias summary. Policy-side quantities were held fixed. The full 2,543,698-paper analysis is shown as a deterministic anchor row rather than as another repeated sample.

Table 13. Sample-stability ladder for the research corpus.

Sample Size	Mean SD of SDG coverage shares	Mean within-SDG semantic gap	Policy-text calibration bias
1,000	0.007	0.458 ± 0.015	0.328 ± 0.005
2,000	0.005	0.434 ± 0.012	0.327 ± 0.003
5,000	0.003	0.418 ± 0.008	0.328 ± 0.002
10,000	0.002	0.410 ± 0.005	0.328 ± 0.001
20,000	0.002	0.405 ± 0.004	0.328 ± 0.001
50,000	0.001	0.402 ± 0.002	0.328 ± 0.001
100,000	0.001	0.401 ± 0.002	0.328 ± 0.000
200,000	0.000	0.401 ± 0.001	0.328 ± 0.000
500,000	0.000	0.400 ± 0.001	0.328 ± 0.000
1,000,000	0.000	0.400 ± 0.000	0.328 ± 0.000
2,000,000	0.000	0.400 ± 0.000	0.328 ± 0.000
Full corpus	–	0.400	0.328

Notes: Mean SD of SDG coverage shares is the unweighted mean of the per-SDG standard deviations in research coverage share across the 100 random draws at each sampled tier. The remaining columns report the mean and draw-to-draw standard deviation of the headline downstream metrics. The full-corpus row is deterministic and therefore has no variance term.

Plain-language guide.[1] The stability ladder shows that the clearest practical plateau appears by roughly 200,000 papers, where the mean semantic gap (∼0.416) and policy–research score gap (∼0.342) already stabilise to within 0.001 of the full-corpus values. From

[1]The three metrics are: (1) mean SD of SDG coverage shares — how much the topic balance moves across repeated samples; (2) mean semantic gap — how different research and policy language are within the same SDG; (3) policy–research score gap — how much the scoring instrument favours policy text over research text regardless of content.

500,000 papers upward, draw-to-draw dispersion is negligible. The full 2,543,698-paper result is retained as the primary estimate because it uses all available data while delivering the tightest values, but the substantive conclusions are not artefacts of sample size choice. The larger corpus also enables per-SDG subgroup detail and tighter rank-order precision that the plateau minimum does not guarantee.

D Embedding-Model Sensitivity

D.1 Rationale

The main-text analysis uses `all-MiniLM-L6-v2` (384-dimensional MiniLM), which was chosen primarily for CPU efficiency as noted in Section 3.4. MiniLM is a lightweight transformer that accomplishes the task at a fraction of the computational cost of larger encoders. However, a concern common to all embedding-based content analysis is whether the central findings are artefacts of the specific encoder rather than properties of the underlying text corpora.

To test this, I repeat the full analysis pipeline with `all-mpnet-base-v2` (768-dimensional MPNet), a different sentence-transformer architecture that has also been benchmarked for SDG-association tasks (Gjorgjevikj et al., 2025). MPNet uses a distinct attention mechanism from MiniLM and operates at twice the embedding dimension, making it a natural sensitivity check: if the main findings replicate under this alternative encoder, they are unlikely to be an artefact of the original model choice.

D.2 Results

The comparison is structured around the three principal outputs of the main-text analysis: the per-SDG validation F1 scores, the SDG coverage gap, and the per-SDG semantic gap values. For each output, I compute the Spearman rank correlation and the Pearson correlation between the two models' results across the 17 SDGs.

Table 14 reports the per-SDG values for both models. The Spearman correlations across all 17 SDGs are reported at the foot of each column pair. Validation F1 scores show the strongest agreement across models, followed by coverage research proportions, while semantic gap rankings are somewhat more variable (as expected given the smaller effective sample size per SDG in the policy corpus). No SDG exhibits a material reversal in rank between the two models for any reported quantity.

These results support the conclusion that the dissertation's principal findings – the per-SDG coverage rankings, the semantic-gap hierarchy, and the coverage–semantic interaction – are not artefacts of the specific embedding encoder. They reflect genuine properties of the

Table 14. Per-SDG comparison of validation F1, coverage gap, and semantic gaps between the canonical MiniLM model and the MPNet sensitivity model. The final row reports Spearman's ρ across the 17 SDGs.

	Validation F1		Coverage gap		Semantic gap	
SDG	MiniLM	MPNet	MiniLM	MPNet	MiniLM	MPNet
SDG 1	0.687 (10)	0.676 (11)	0.0046 (17)	0.0061 (17)	0.253 (15)	0.420 (7)
SDG 2	0.854 (5)	0.848 (5)	0.0328 (13)	0.0306 (10)	0.324 (11)	0.345 (11)
SDG 3	0.680 (13)	0.630 (14)	0.1850 (1)	0.2172 (1)	0.423 (10)	0.452 (4)
SDG 4	0.907 (1)	0.918 (1)	0.1398 (2)	0.1597 (2)	0.287 (13)	0.263 (14)
SDG 5	0.737 (7)	0.667 (12)	0.0211 (14)	0.0230 (12)	0.249 (16)	0.238 (17)
SDG 6	0.878 (2)	0.857 (3)	0.0357 (12)	0.0258 (11)	0.491 (4)	0.433 (6)
SDG 7	0.878 (2)	0.885 (2)	0.0781 (4)	0.0786 (4)	0.497 (3)	0.522 (1)
SDG 8	0.548 (15)	0.554 (15)	0.0165 (15)	0.0182 (14)	0.269 (14)	0.251 (16)
SDG 9	0.678 (14)	0.727 (8)	0.0972 (3)	0.1428 (3)	0.234 (17)	0.258 (15)
SDG 10	0.733 (8)	0.655 (13)	0.0080 (16)	0.0062 (16)	0.432 (8)	0.350 (10)
SDG 11	0.545 (16)	0.549 (16)	0.0537 (8)	0.0771 (5)	0.447 (7)	0.479 (3)
SDG 12	0.684 (11)	0.725 (9)	0.0358 (11)	0.0378 (8)	0.424 (9)	0.408 (8)
SDG 13	0.694 (9)	0.693 (10)	0.0435 (9)	0.0200 (13)	0.528 (2)	0.406 (9)
SDG 14	0.822 (6)	0.811 (6)	0.0390 (10)	0.0344 (9)	0.485 (5)	0.440 (5)
SDG 15	0.683 (12)	0.753 (7)	0.0690 (6)	0.0395 (7)	0.465 (6)	0.334 (12)
SDG 16	0.870 (4)	0.853 (4)	0.0743 (5)	0.0748 (6)	0.323 (12)	0.281 (13)
SDG 17	0.300 (17)	0.302 (17)	0.0658 (7)	0.0081 (15)	0.671 (1)	0.518 (2)
Spearman ρ	0.8547		0.8578		0.7230	

Note. Rank 1 = highest F1 / highest coverage proportion / largest semantic gap per model.

Notes: Rank 1 = highest F1 / highest coverage proportion / largest gap per model.

research and policy corpora rather than noise introduced by the measurement instrument.

E Register-Adjustment Robustness and Diagnostics

This appendix reports exploratory register-adjustment procedures that probe whether broad research-policy register differences contribute to the raw semantic gap. Following Biber and Conrad's distinction between register, genre, and style, I treat the research-policy contrast primarily as a register contrast: the corpora differ in situation of use, audience, institutional setting, and communicative purpose, not merely in surface wording (Biber & Conrad, 2009).

The logic of this appendix is informed by multidimensional register analysis: register differences are structured patterns of co-occurring linguistic features — not merely surface decoration around stable content (Biber, 1988). A learned research-policy direction in embedding space may therefore capture a mixture of modality, abstraction, institutional stance, and substantive theme. This is why the adjusted results are reported as robustness diagnostics rather than corrected estimates.

E.1 One-Direction Global Sensitivity Check

The simplest robustness step estimates one broad research-versus-policy direction in the shared SBERT space using a balanced logistic-regression classifier trained on held-out splits. On the held-out test set, this classifier achieves accuracy 0.981, ROC-AUC (Receiver Operating Characteristic Area Under the Curve — a classification performance metric) 0.998, F1 0.981, precision 0.978, and recall 0.984 using 20,000 sampled texts per class before the train/validation/test split. This shows that a strong corpus-level research–policy axis exists.

Formally, let $\mathbf{x}$ be a unit-normalised embedding and let $\mathbf{g}$ denote the learned global research–policy direction. The one-direction adjustment first subtracts the orthogonal projection of $\mathbf{x}$ onto $\mathbf{g}$,

$$\mathbf{x}^{\perp} = \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{g}}{\|\mathbf{g}\|^2}\mathbf{g},$$

and then renormalises the residual to unit length before centroid averaging. I describe the result as *register-adjusted*, not "debiased": the subtraction removes one estimated global corpus axis, not every possible register or corpus-composition difference.

Subtracting that one global direction reduces the mean gap from 0.400 to 0.374, a change of -0.026. The largest shrinkage appears in SDG 7 (-0.036), while the largest remaining adjusted gap is still SDG 17 at 0.655. The substantive interpretation is limited: this subtraction attenuates one broad corpus-level axis, but it does not cleanly identify a pure substantive gap net of all register effects.

Table 15. Global one-direction register sensitivity check for the semantic gap.

SDG	Description	Raw gap	Adjusted gap	Δ gap	n_{res}	$n_{pol\ docs}$
SDG 17	Partnerships for the Goals	0.671	0.655	-0.017	167,440	1,709
SDG 13	Climate Action	0.528	0.505	-0.023	110,645	1,503
SDG 6	Clean Water and Sanitation	0.491	0.463	-0.028	90,715	302
SDG 14	Life Below Water	0.485	0.462	-0.023	99,162	343
SDG 7	Affordable and Clean Energy	0.497	0.462	-0.036	198,743	367
SDG 15	Life on Land	0.465	0.441	-0.024	175,447	353
SDG 11	Sustainable Cities and Communities	0.447	0.418	-0.029	136,616	297
SDG 10	Reduced Inequalities	0.432	0.411	-0.021	20,323	208
SDG 12	Responsible Consumption and Production	0.424	0.391	-0.033	90,960	286
SDG 3	Good Health and Well-Being	0.423	0.391	-0.032	470,627	391
SDG 2	Zero Hunger	0.324	0.300	-0.024	83,441	347
SDG 16	Peace, Justice and Strong Institutions	0.323	0.297	-0.026	189,046	1,183
SDG 4	Quality Education	0.287	0.254	-0.033	355,720	366
SDG 8	Decent Work and Economic Growth	0.269	0.241	-0.028	41,898	312
SDG 1	No Poverty	0.253	0.231	-0.021	11,804	351
SDG 5	Gender Equality	0.249	0.228	-0.021	53,741	499
SDG 9	Industry, Innovation and Infrastructure	0.234	0.207	-0.027	247,370	408
Mean gap		0.400	0.374	-0.026		

Notes: Raw gap = the main semantic-gap estimate from the dissertation. Adjusted gap = the same within-SDG gap after subtracting one learned global research–policy direction from each embedding.

E.2 Additional Confidence Checks on Global Subtraction

The one-direction subtraction is itself testable. The adjusted re-separability audit shows that research and policy remain highly separable even after this first direction is removed: the raw held-out ROC-AUC is 0.998, while the adjusted held-out ROC-AUC remains 0.996 (relative change -0.003). Held-out-SDG generalization is also strong, with mean held-out ROC-AUC 0.997 and mean held-out F1 0.977, which indicates that the learned axis is broad rather than purely SDG-specific leakage.

At the same time, the multi-direction subtraction curve shows that the mean adjusted gap continues to decline from 0.374 at $k = 1$ to 0.310 at $k = 5$, while the residual classifier ROC-AUC at $k = 5$ is still 0.964. The topic-matched nearest-neighbor check also remains far from a collapse, moving from 0.425 in raw space to 0.430 after subtraction. The one-direction residual cannot be treated as pure substantive divergence; it is better understood as a limited sensitivity analysis against one strong global register axis.

E.3 SDG-Balanced Global Controls and Within-SDG Stress Tests

The global one-direction subtraction may still reflect SDG composition in the sampled classifier data. To probe that, I add an SDG-balanced global classifier that draws equal numbers of research and policy texts from every SDG before learning one shared research–policy direction. This is a stronger global sensitivity check because it targets broad

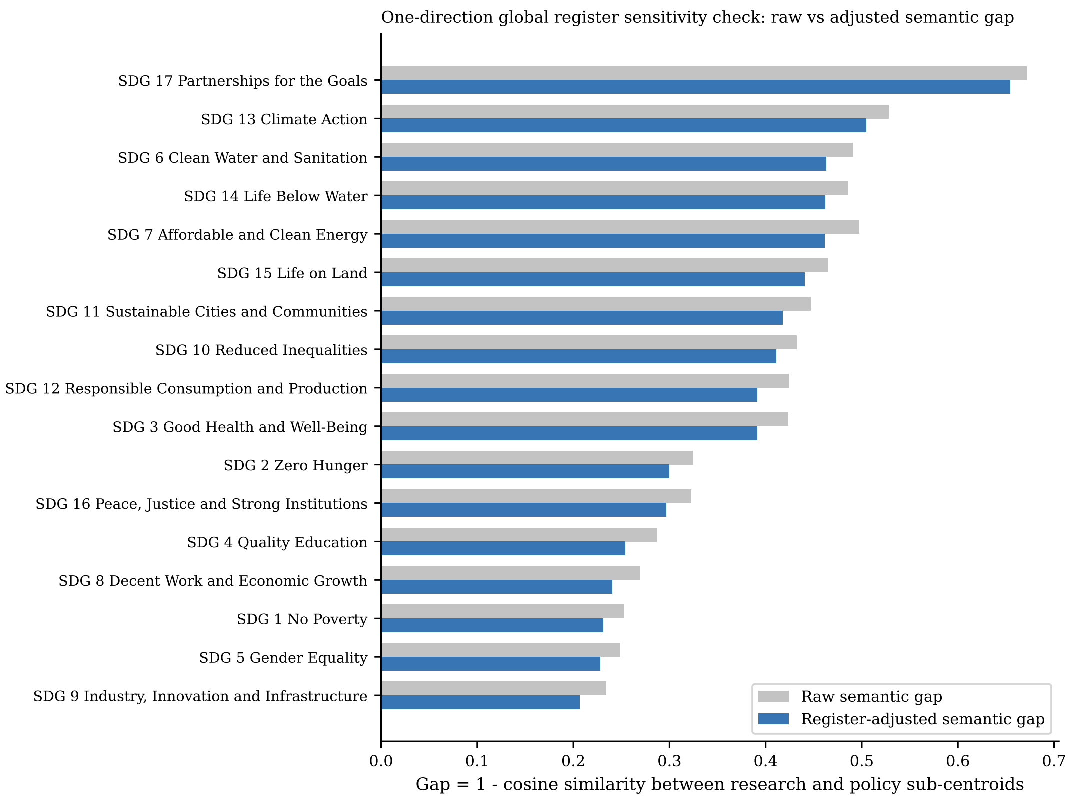


Figure 7. Global one-direction register sensitivity check: raw versus adjusted semantic gap by SDG. Raw gap is shown in grey and the one-direction adjusted gap in blue.

corpus-level register while controlling SDG composition. In the current run, the SDG-balanced classifier remains strongly separable (accuracy 0.974, macro-F1 0.974, ROC-AUC 0.996), and the mean gap becomes 0.332.

I also fit within-SDG classifiers, but these are interpreted differently. Because each classifier is estimated inside a single SDG, subtracting the resulting direction risks removing the very within-goal research–policy contrast the dissertation is trying to measure. Their mean adjusted gap (0.227) is therefore treated as a stress-test result rather than as a better estimate. The cosine similarity between the SDG-balanced global direction and the average within-SDG direction is 0.914, which helps show whether the SDG-specific directions mostly line up with the broad global axis or vary goal by goal.

E.4 Interpreting the Learned Register Direction

The previous appendix checks establish that a strong research–policy direction exists in the embedding space and that its subtraction changes semantic-gap magnitudes. A separate question is what that learned direction is actually capturing. Two descriptive diagnostics help answer that. First, every text can be projected onto the SDG-balanced global register vector, producing ranked lists of the most policy-like and most research-like texts without fitting any new model. Second, the same register direction can be compared directly with

Table 16. Additional checks on the one-direction global register subtraction.

Check	Headline estimate	Interpretation
Adjusted re-separability	Raw ROC-AUC 0.998; adjusted 0.996(Δ -0.003)	One removed direction leaves cross-corpus separability very high.
Held-out SDG generalization	Mean held-out ROC-AUC 0.997; mean F1 0.977	The learned register axis transfers across unseen SDG blocks, so it is not merely within-SDG topic leakage.
Multi-direction subtraction	Mean adjusted gap 0.374 at $k = 1$ and 0.310 at $k = 5$	Additional discriminative directions continue to compress gap magnitude, although the largest-gap SDGs remain concentrated in SDG 6, SDG 13, SDG 14, and SDG 17.
Topic-matched nearest neighbor	Mean matched gap 0.425 raw and 0.430 adjusted (Δ 0.005)	Within-SDG nearest cross-corpus matches remain far apart after subtraction and become slightly less similar on average.

Notes: These diagnostics test whether the first subtraction plausibly attenuates a broad corpus-level register axis without claiming that the residual is a fully purified substantive gap.

the canonical OSDG-derived SDG centroids. If the direction were close to orthogonal to those centroids, it would look more like a broad wrapper-language axis; if it aligns materially with SDG centroids, it likely mixes register and substantive thematic structure.

The projection examples are included for manual inspection rather than automatic summarization. In the current run, the mean score of the top policy-like examples is 0.460, while the mean score of the bottom research-like examples is -0.235. The SDG-centroid comparison is more formal: the mean absolute cosine between the global register vector and the 17 SDG centroids is 0.372, the median is 0.382, and the maximum is 0.417. The strongest alignment occurs at SDG 17, while the weakest occurs at SDG 16. These values indicate overlap with SDG semantic structure rather than a clean separation of style from substance.

E.5 Regression Decomposition of the Embedding Space

The classifier-based register direction is discriminative by construction, so a final concern is that its overlap with SDG semantic structure might partly be an artefact of optimizing corpus separation. To probe that possibility, I add a Rodriguez-style embedding regression diagnostic. Each embedding coordinate is treated as an outcome in a linear model with corpus register, SDG fixed effects, and register–SDG interactions. This does not optimize a classifier; it decomposes embedding variation associated with register after controlling for SDG structure.

In the current run, the cosine between the regression-derived global register vector and the SDG-balanced classifier vector is 0.631, which falls in the partial agreement range. The

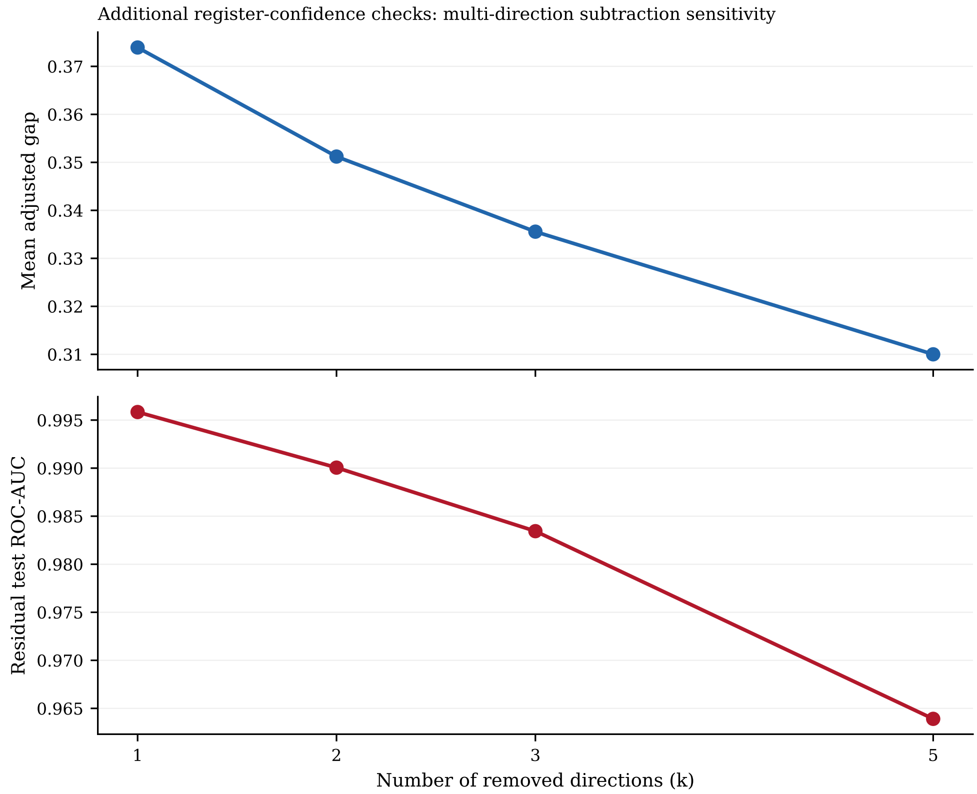


Figure 8. Confidence checks on global register subtraction. The top panel shows how the mean adjusted semantic gap changes as more discriminative research–policy directions are removed. The bottom panel shows the residual held-out classifier ROC-AUC after the same sequence of removals.

regression-derived global register vector remains materially aligned with SDG centroids: mean absolute cosine 0.520, median 0.516, maximum 0.719, strongest alignment SDG 1, weakest SDG 16. The mean regression-global adjusted gap is 0.251, while the mean regression within-SDG adjusted gap is 0.012. The latter is interpreted as an especially aggressive over-subtraction stress test because the within-SDG regression vectors are very close to estimating the within-goal policy–research contrast itself.

Table 17. SDG-aware register robustness hierarchy.

Method	Headline estimate	Interpretation
SDG-balanced global classifier	Test accuracy 0.974; macro-F1 0.974; ROC-AUC 0.996; mean adjusted gap 0.332	One global research-policy direction is learned after equalising the SDG composition of the training sample.
Within-SDG classifiers	Mean test accuracy 0.974; mean macro-F1 0.974; mean ROC-AUC 0.996; mean adjusted gap 0.227	SDG-specific register directions are fit where sufficient data exist (17 SDGs fit, 0 skipped).
Direction stability	Global-vs-average cosine 0.914	Higher cosine similarity means the within-SDG register directions point in roughly the same direction as the SDG-balanced global control.

Notes: The SDG-balanced classifier is treated as a stronger global sensitivity check. The within-SDG design is treated as an over-subtraction stress test because it learns and removes within-goal research–policy contrasts directly.

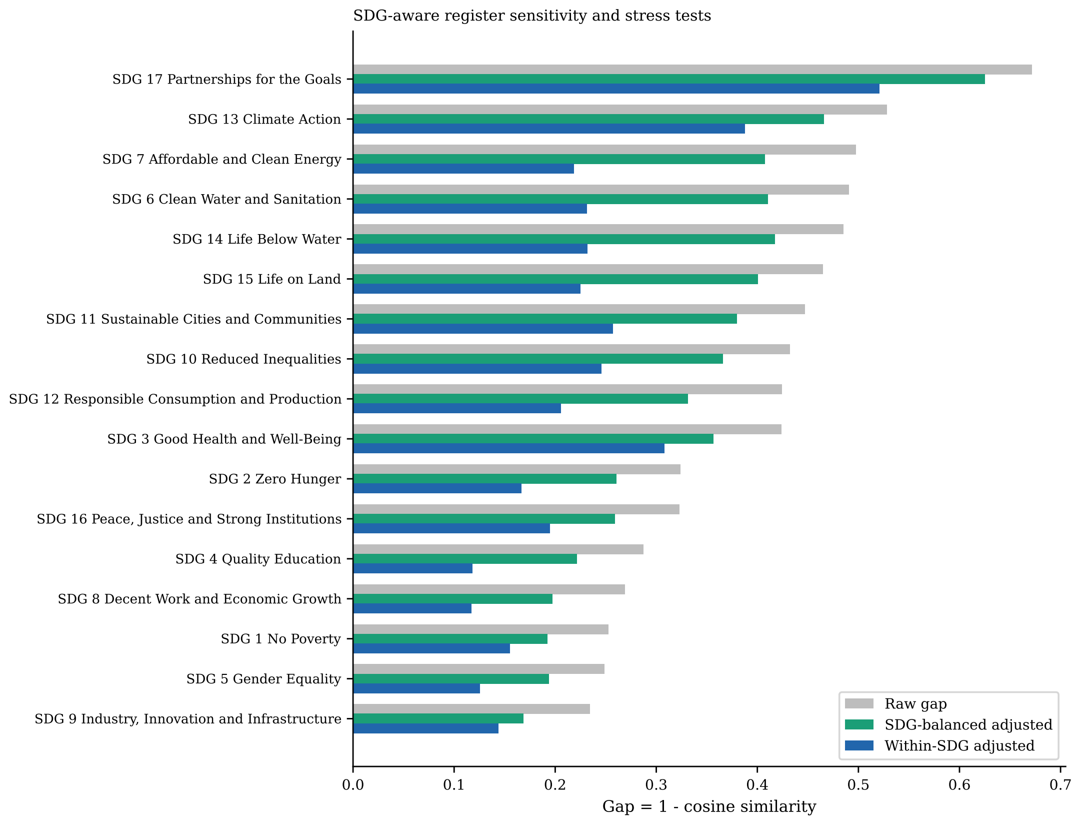


Figure 9. Raw semantic gap compared with SDG-aware register procedures. The green bars show the SDG-balanced global sensitivity check. The blue bars show the within-SDG stress test, which is more prone to over-subtraction.

Table 18. Alignment between the SDG-balanced global register direction and the canonical SDG centroids.

SDG	$\cos(g, c_k)$	$\lvert \cos(g, c_k) \rvert$	$\cos(g_k, c_k)$
SDG 1	0.347	0.347	0.433
SDG 2	0.378	0.378	0.558
SDG 3	0.381	0.381	0.491
SDG 4	0.347	0.347	0.462
SDG 5	0.335	0.335	0.466
SDG 6	0.393	0.393	0.569
SDG 7	0.374	0.374	0.575
SDG 8	0.409	0.409	0.474
SDG 9	0.408	0.408	0.520
SDG 10	0.345	0.345	0.523
SDG 11	0.411	0.411	0.450
SDG 12	0.386	0.386	0.480
SDG 13	0.382	0.382	0.519
SDG 14	0.382	0.382	0.588
SDG 15	0.390	0.390	0.524
SDG 16	0.240	0.240	0.263
SDG 17	0.417	0.417	0.551

Notes: The final column reports the matching within-SDG register-versus-centroid cosine when an SDG-specific register vector is available.

Table 19. Manual-inspection examples from the learned global register direction. The policy-like rows have the highest positive projection onto the global register vector; the research-like rows have the lowest projection.

Side	SDG	Score	ID	Preview
Policy-like	SDG 3	0.486	sdgi_00878_3	3.b. The country has seen an increase in total net official development assistance for medical resea
Policy-like	SDG 13	0.473	sdgi_05080_12	It is a cause of concern that net ODA flows slid down by 4.3 per cent in 2018, with a dwindling shar
Policy-like	SDG 5	0.471	sdgi_01520_21	When taking into account also other official flows, private funds mobilized through public intervent
Policy-like	SDG 3	0.470	sdgi_00810_14	Modest progress is being registered towards achievement of UHC, as illustrated by improvements on th
Policy-like	SDG 17	0.469	sdgi_01173_15	64 EU external action (10) See UN SDG Report 2021 (page 34) and UN SDG Report 2022 (Page 33). (11)
Research-like	SDG 14	-0.261	W4389228499	Research on the applicability of regularization methods in ship magnetic field modeling based on mag
Research-like	SDG 7	-0.256	W4404845190	A fast Time-domain identification of bushing dynamics. The identification of bushings dynamics is cr
Research-like	SDG 16	-0.248	W3134161550	Review on Aluminum and Steel Semi-rigid Connections Behavior Design Model. Several research reports
Research-like	SDG 7	-0.239	W4376643972	Sparse identification of Lagrangian for nonlinear dynamical systems via proximal gradient method. Th
Research-like	SDG 7	-0.239	W4400996301	Research on Stock Price Prediction Model based on Clustering-Sliced LDA and Gaussian Process Regress

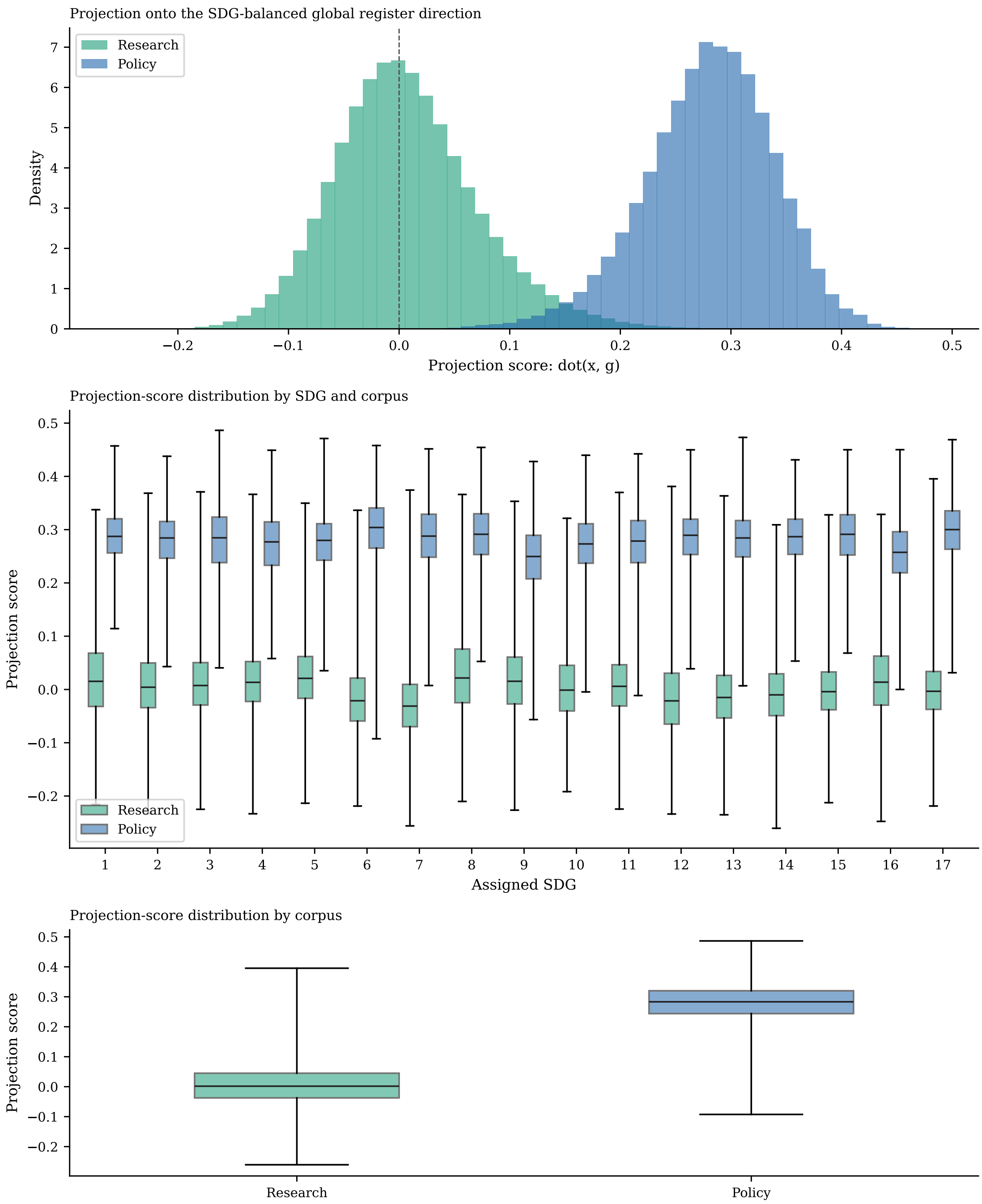


Figure 10. Distribution of projection scores onto the SDG-balanced global register direction. *Notes:* The figure is descriptive only: it exposes what kinds of texts align most strongly with the learned global axis.

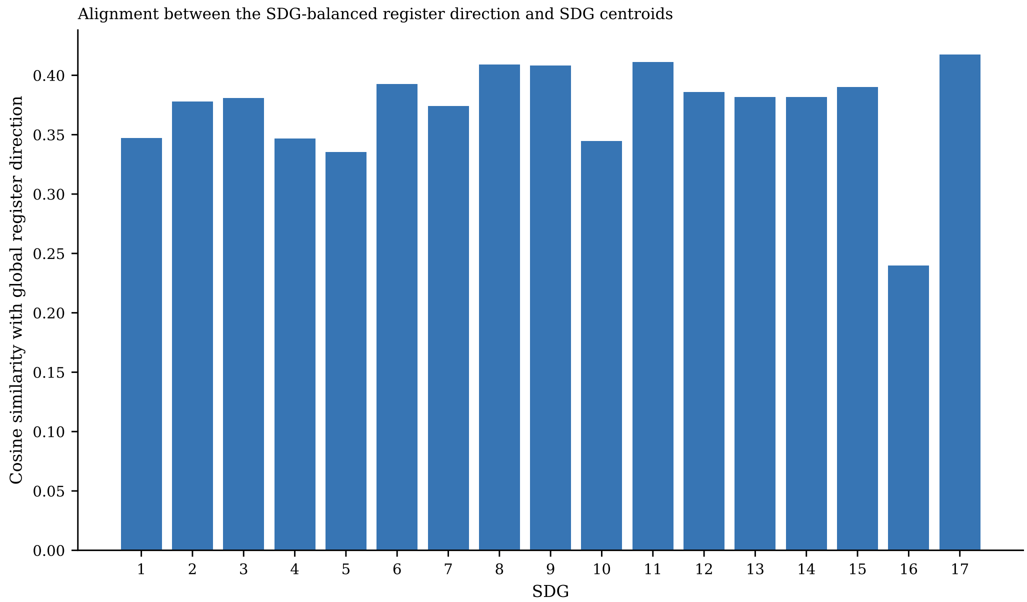


Figure 11. Signed cosine similarity between the SDG-balanced global register direction and each OSDG-derived SDG centroid. Larger magnitudes indicate stronger overlap between the learned register direction and a specific SDG axis.

Table 20. Regression-derived versus classifier-derived register alignment with the canonical SDG centroids.

SDG	$\cos(g_{\mathrm{clf}}, c_k)$	$\cos(g_{\mathrm{reg}}, c_k)$	$\cos(g_{\mathrm{reg},k}, c_k)$
SDG 1	0.347	0.719	0.719
SDG 2	0.378	0.613	0.714
SDG 3	0.381	0.531	0.705
SDG 4	0.347	0.461	0.538
SDG 5	0.335	0.516	0.684
SDG 6	0.393	0.475	0.741
SDG 7	0.374	0.478	0.730
SDG 8	0.409	0.639	0.622
SDG 9	0.408	0.519	0.718
SDG 10	0.345	0.651	0.705
SDG 11	0.411	0.543	0.647
SDG 12	0.386	0.477	0.710
SDG 13	0.382	0.503	0.732
SDG 14	0.382	0.388	0.805
SDG 15	0.390	0.452	0.719
SDG 16	0.240	0.343	0.545
SDG 17	0.417	0.526	0.767

Notes: The second column reports the SDG-balanced classifier direction, the third reports the regression-derived global register direction, and the fourth reports SDG-specific regression register directions.

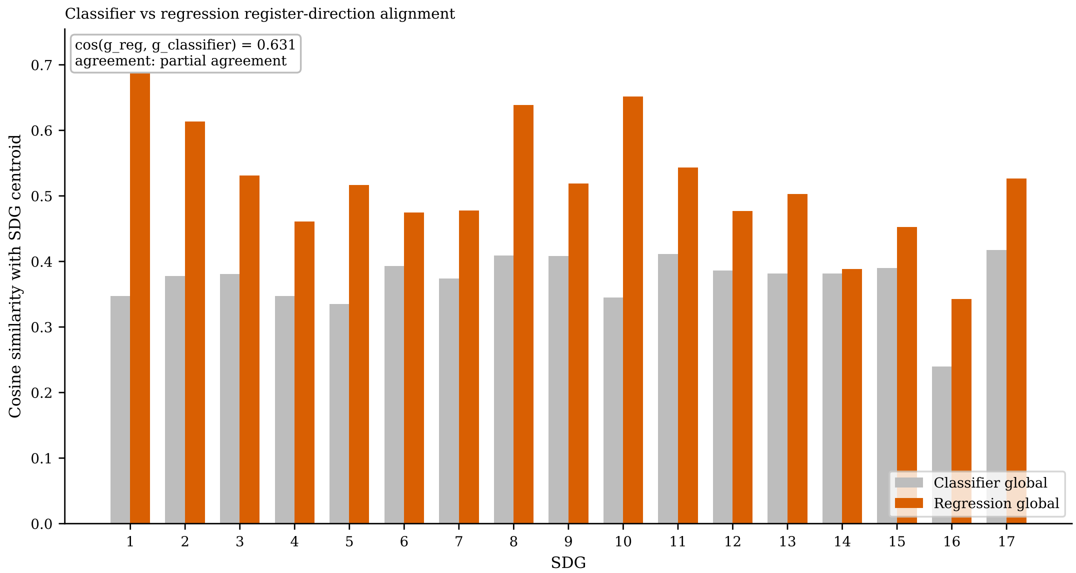


Figure 12. Classifier-derived versus regression-derived register alignment with the OSDG centroids.

Notes: The annotation reports the cosine similarity between the global regression vector and the SDG-balanced classifier vector.

F Declaration of AI Use

In accordance with the University's guidance on generative AI ("Use of GenAI is permitted to support assignments"), this appendix declares the use of AI tools throughout the dissertation. All use was conducted with human oversight and control; AI-generated outputs were reviewed, modified, and independently verified before inclusion. No AI-generated text was submitted without such verification. This declaration follows the order of permitted activities specified in the assessment guidelines. AI-assisted activities not explicitly listed (e.g., code debugging, plotting, LaTeX troubleshooting, literature summarisation) fall under the same oversight and verification standards described in the relevant rows above.

Table 21. Declaration of AI use by permitted activity.

Activity	**AI tool(s)**	**How output was used**
Researching and refining ideas	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	ChatGPT was used iteratively to develop ideas that aligned with the author's intent. Coding agents independently checked data and code availability to confirm feasibility. All final decisions and conclusions are the author's own. Final research design was formally presented to and approved by supervisor.
Information retrieval / background research	Gemini, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	AI tools read the manuscript for context and recommended papers to investigate further. All cited sources were independently verified against the original publications.
Drafting an outline / organising thoughts	MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Generated code scaffolding and assisted with the full analysis pipeline (embedding, scoring, validation, robustness, appendix outputs). The centroid-assignment step was verified against hand-labeled samples, and the scoring and coverage-gap functions were unit-tested. Analytical decisions were directed by the author.
Refining research questions	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Overlaps with row 1. ChatGPT sharpened framing based on reviewed literature; coding agents confirmed questions were answerable with available data and models. The author retained full control over the final formulation. RQs presented to and approved by supervisor.
Checking spelling and grammar	ChatGPT, Gemini, Claude (web/mobile), MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Coding agents provided inline stylistic and phrasing suggestions during drafting. Web and mobile versions of ChatGPT, Gemini, and Claude were used for editorial review and proofreading. All suggestions were accepted or rejected per standard editorial judgment.